

Geometry 4

Problem 1 (Geodesics in $O(n)$ via matrix exponentials).

Consider the standard (Euclidean) inner product on the space $M(n)$ of real $n \times n$ matrices,

$$\langle L, K \rangle = \text{Tr}(LK^t),$$

and the induced Riemannian metric on the submanifold $X = O(n) \subset M(n)$. For $A \in T_I X$, consider the curve $\alpha : \mathbb{R} \rightarrow M(n)$ defined by

$$\alpha(t) = \exp(tA).$$

Prove that α is a curve in X and that it is a geodesic of X , by showing that for all $t \in \mathbb{R}$,

$$\alpha''(t) = A^2\alpha(t) \quad \text{is orthogonal to} \quad T_{\alpha(t)}X.$$

Solution. Let $M(n)$ be the space of real $n \times n$ matrices with inner product

$$\langle L, K \rangle = \text{Tr}(LK^t).$$

Let $X = O(n) \subset M(n)$. The tangent space at the identity is

$$T_I O(n) = \{A \in M(n) : A^t = -A\}$$

(skew-symmetric matrices). Fix $A \in T_I O(n)$ and define

$$\alpha(t) = \exp(tA).$$

Step 1: $\alpha(t) \in O(n)$ for all t . Using $(e^B)^t = e^{Bt}$ and $A^t = -A$,

$$\alpha(t)^t \alpha(t) = e^{tA^t} e^{tA} = e^{-tA} e^{tA} = I,$$

so $\alpha(t) \in O(n)$ for all $t \in \mathbb{R}$. Hence α is a curve in X .

Step 2: compute $\alpha'(t)$ and $\alpha''(t)$. Since $\frac{d}{dt}e^{tA} = Ae^{tA}$,

$$\alpha'(t) = A\alpha(t), \quad \alpha''(t) = A^2\alpha(t).$$

Step 3: describe the tangent space $T_Q O(n)$ and show orthogonality. For $Q \in O(n)$,

$$T_Q O(n) = \{BQ : B^t = -B\}.$$

(Indeed, differentiating $Q(t)^t Q(t) = I$ at $t = 0$ gives $Q^t V + V^t Q = 0$, i.e. $V = BQ$ with B skew.)

Now fix t and write $Q = \alpha(t)$. For any tangent vector $V = BQ \in T_Q O(n)$ with $B^t = -B$,

$$\langle \alpha''(t), V \rangle = \langle A^2 Q, BQ \rangle = \text{Tr}((A^2 Q)^t (BQ)) = \text{Tr}(Q^t (A^2)^t BQ).$$

Since $A^t = -A$, we have $(A^2)^t = (A^t)^2 = (-A)^2 = A^2$, so A^2 is symmetric. Using cyclicity of the trace and $QQ^t = I$,

$$\text{Tr}(Q^t A^2 BQ) = \text{Tr}(A^2 BQQ^t) = \text{Tr}(A^2 B).$$

But A^2 is symmetric and B is skew-symmetric, hence

$$\text{Tr}(A^2 B) = \text{Tr}((A^2 B)^t) = \text{Tr}(B^t (A^2)^t) = \text{Tr}((-B)A^2) = -\text{Tr}(A^2 B),$$

so $\text{Tr}(A^2 B) = 0$. Therefore

$$\langle \alpha''(t), V \rangle = 0 \quad \text{for all } V \in T_{\alpha(t)} O(n),$$

i.e. $\alpha''(t) = A^2 \alpha(t)$ is orthogonal to $T_{\alpha(t)} X$ for all t .

Step 4: geodesic conclusion. Because $O(n) \subset (M(n), \langle \cdot, \cdot \rangle)$ is a submanifold of a Euclidean space, a curve in $O(n)$ is a geodesic (for the induced metric) iff its ambient acceleration is normal to the submanifold:

$$\alpha \text{ is geodesic} \iff \alpha''(t) \perp T_{\alpha(t)} O(n) \quad \forall t.$$

Thus $\alpha(t) = e^{tA}$ is a geodesic in $O(n)$.

Theory (for Problem 1: geodesics on $O(n) \subset M(n)$)

Let $M(n)$ denote the vector space of real $n \times n$ matrices, equipped with the (Euclidean) inner product

$$\langle L, K \rangle = \text{Tr}(LK^t).$$

This identifies $M(n) \cong \mathbb{R}^{n^2}$ as a Euclidean space. The orthogonal group

$$O(n) = \{Q \in M(n) : Q^t Q = I\}$$

is a smooth submanifold of $M(n)$, and it inherits a Riemannian metric by restricting $\langle \cdot, \cdot \rangle$ to each tangent space.

(1) Tangent space of $O(n)$. Let $Q \in O(n)$, and let $V \in T_Q O(n)$. Choose a smooth curve $Q(t) \in O(n)$ with $Q(0) = Q$ and $Q'(0) = V$. Differentiating the constraint $Q(t)^t Q(t) = I$ at $t = 0$ gives

$$V^t Q + Q^t V = 0.$$

Equivalently, $Q^t V$ is skew-symmetric. Writing $B := VQ^t$, this condition becomes $B^t = -B$, and $V = BQ$. Hence

$$T_Q O(n) = \{BQ : B^t = -B\}.$$

In particular,

$$T_I O(n) = \{A \in M(n) : A^t = -A\} = \mathfrak{so}(n).$$

(2) Symmetric matrices are orthogonal to skew-symmetric matrices. If $S^t = S$ and $B^t = -B$, then

$$\operatorname{Tr}(SB) = \operatorname{Tr}((SB)^t) = \operatorname{Tr}(B^t S^t) = \operatorname{Tr}((-B)S) = -\operatorname{Tr}(SB),$$

so $\operatorname{Tr}(SB) = 0$. Thus symmetric and skew-symmetric matrices are orthogonal with respect to $\langle L, K \rangle = \operatorname{Tr}(LK^t)$.

(3) Geodesics in an isometrically embedded submanifold. Let $N \subset \mathbb{R}^m$ be a submanifold with the metric induced from the ambient Euclidean inner product. A smooth curve $\gamma(t) \in N$ is a geodesic in N if and only if its ambient acceleration has no tangential component, i.e.

$$\gamma \text{ is a geodesic} \iff \gamma''(t) \perp T_{\gamma(t)}N \text{ for all } t.$$

Equivalently, γ is geodesic precisely when $\gamma''(t)$ is a normal vector to N along the curve.

(4) Matrix exponential curves. For any $A \in M(n)$,

$$\frac{d}{dt}e^{tA} = Ae^{tA}.$$

If $A^t = -A$, then $(e^{tA})^t = e^{tA^t} = e^{-tA}$, hence

$$(e^{tA})^t e^{tA} = e^{-tA} e^{tA} = I,$$

so $e^{tA} \in O(n)$ for all $t \in \mathbb{R}$.

Problem 2 (Curvature from the length of small geodesic circles).

Using geodesic polar coordinates, show that for a surface S and any point $p \in S$, the Gaussian curvature can be expressed as

$$K(p) = \lim_{r \rightarrow 0} \frac{3(2\pi r - L(r))}{\pi r^3},$$

where $L(r)$ denotes the length of the geodesic circle of radius r centered at p . [*Hint: Taylor expansion.*]

Solution. Fix $p \in S$. In *geodesic polar coordinates* (r, θ) around p , the metric has the form

$$ds^2 = dr^2 + G(r, \theta)^2 d\theta^2,$$

with $G(0, \theta) = 0$ and $G_r(0, \theta) = 1$. The geodesic circle of radius r is $\{r = \text{const}\}$, and its length is

$$L(r) = \int_0^{2\pi} \sqrt{g_{\theta\theta}} d\theta = \int_0^{2\pi} G(r, \theta) d\theta.$$

A standard fact from Jacobi field theory is that G satisfies (along each fixed θ)

$$G_{rr}(r, \theta) + K(r, \theta) G(r, \theta) = 0,$$

where $K(r, \theta)$ is the Gaussian curvature at the point $\exp_p(r e_\theta)$. Since $K(r, \theta) = K(p) + O(r)$ as $r \rightarrow 0$, a Taylor expansion using

$$G(0, \theta) = 0, \quad G_r(0, \theta) = 1, \quad G_{rr}(0, \theta) = 0, \quad G_{rrr}(0, \theta) = -K(p)$$

gives

$$G(r, \theta) = r - \frac{K(p)}{6} r^3 + o(r^3) \quad (r \rightarrow 0),$$

uniformly in θ . Integrating over θ ,

$$L(r) = \int_0^{2\pi} \left(r - \frac{K(p)}{6} r^3 + o(r^3) \right) d\theta = 2\pi r - \frac{K(p)}{6} 2\pi r^3 + o(r^3).$$

Hence

$$2\pi r - L(r) = \frac{K(p)}{6} 2\pi r^3 + o(r^3) = \frac{K(p)\pi}{3} r^3 + o(r^3),$$

so

$$\lim_{r \rightarrow 0} \frac{3(2\pi r - L(r))}{\pi r^3} = \lim_{r \rightarrow 0} \frac{3 \left(\frac{K(p)\pi}{3} r^3 + o(r^3) \right)}{\pi r^3} = K(p),$$

which is exactly the desired formula.

Theory (for Problem 2: curvature from the length of small geodesic circles)

Let S be a smooth surface with Riemannian metric ds^2 , and fix a point $p \in S$. For sufficiently small $r > 0$, the exponential map $\exp_p$ is a diffeomorphism from a disk in $T_p S$ onto a neighborhood of p . Using $\exp_p$, one introduces *geodesic polar coordinates* (r, θ) around p , where r is the geodesic distance from p and θ is the direction in $T_p S$.

(1) Polar form of the metric. In geodesic polar coordinates, the metric takes the form

$$ds^2 = dr^2 + G(r, \theta)^2 d\theta^2,$$

where $G(r, \theta) > 0$ for $r > 0$, and the initial conditions at $r = 0$ are

$$G(0, \theta) = 0, \quad G_r(0, \theta) = 1.$$

Geometrically, $G(r, \theta)$ is the length scale factor in the angular direction.

(2) Length of a geodesic circle. The geodesic circle of radius r centered at p is the curve $\{r = \text{const}\}$. Since $g_{\theta\theta} = G(r, \theta)^2$, its length is

$$L(r) = \int_0^{2\pi} \sqrt{g_{\theta\theta}} d\theta = \int_0^{2\pi} G(r, \theta) d\theta.$$

(3) Jacobi equation for G . For each fixed θ , the function $r \mapsto G(r, \theta)$ satisfies the scalar Jacobi equation

$$G_{rr}(r, \theta) + K(r, \theta) G(r, \theta) = 0,$$

where $K(r, \theta)$ is the Gaussian curvature at the point $\exp_p(r e_\theta)$. This follows by considering the variation of radial geodesics with respect to θ and the associated Jacobi field.

(4) Small- r expansion and curvature. Since $K(r, \theta) = K(p) + O(r)$ as $r \rightarrow 0$, the Jacobi equation and the initial data imply a Taylor expansion of the form

$$G(r, \theta) = r - \frac{K(p)}{6} r^3 + o(r^3) \quad (r \rightarrow 0),$$

uniformly in θ . Integrating this expansion over θ yields the standard asymptotic formula for the length of a small geodesic circle:

$$L(r) = 2\pi r - \frac{K(p)}{6} 2\pi r^3 + o(r^3) = 2\pi r - \frac{K(p)\pi}{3} r^3 + o(r^3).$$

Rearranging gives the curvature-recovery limit used in Problem 2:

$$K(p) = \lim_{r \rightarrow 0} \frac{3(2\pi r - L(r))}{\pi r^3}.$$

Problem 3 (Geodesic curvature of geodesic circles and parallels).

Prove that on a surface of constant Gaussian curvature, geodesic circles have constant geodesic curvature. Suppose that on a surface S there is a point P with the property that locally around P the Gaussian curvature is constant along each geodesic circle centered at P ; show that the geodesic curvature is also constant along these geodesic circles. Finally, find the geodesic curvature of a parallel of latitude on the 2-sphere.

Solution. (A) *Constant Gaussian curvature $\Rightarrow$ geodesic circles have constant geodesic curvature.*

In geodesic polar coordinates around a point p , the metric is

$$ds^2 = dr^2 + G(r, \theta)^2 d\theta^2.$$

The circle $r = r_0$ has tangent vector ∂_θ of length $G(r_0, \theta)$, hence unit tangent

$$T = \frac{1}{G(r_0, \theta)} \partial_\theta.$$

A direct Christoffel-symbol computation for this metric gives

$$\Gamma_{\theta\theta}^r = -G G_r.$$

For the curve $r = r_0$ parametrized by arc length s , we have $ds = G(r_0, \theta) d\theta$, so $\frac{d\theta}{ds} = \frac{1}{G}$. Therefore the covariant acceleration has radial component

$$(\nabla_T T)^r = \Gamma_{\theta\theta}^r \left(\frac{d\theta}{ds} \right)^2 = (-G G_r) \left(\frac{1}{G^2} \right) = -\frac{G_r}{G},$$

and its norm is

$$k_g(r_0, \theta) = \|\nabla_T T\| = \left| \frac{G_r(r_0, \theta)}{G(r_0, \theta)} \right|.$$

So the geodesic curvature is constant along the circle precisely when $G(r_0, \theta)$ is independent of θ .

If the Gaussian curvature is constant $K \equiv K_0$, then the Jacobi equation becomes

$$G_{rr} + K_0 G = 0, \quad G(0, \theta) = 0, \quad G_r(0, \theta) = 1,$$

which has the θ -independent solution $G(r, \theta) = S_{K_0}(r)$. Hence $k_g(r_0, \theta) = |S'_{K_0}(r_0)/S_{K_0}(r_0)|$ is constant along each geodesic circle.

(B) If locally K is constant along each geodesic circle around P , then the geodesic curvature is constant along those circles." " K constant along each geodesic circle" means that in polar coordinates centered at P ,

$$K(r, \theta) = K(r) \quad (\text{no dependence on } \theta).$$

Then for each fixed θ , $G(r, \theta)$ solves the same ODE

$$G_{rr}(r, \theta) + K(r) G(r, \theta) = 0, \quad G(0, \theta) = 0, \quad G_r(0, \theta) = 1,$$

with the same initial conditions. By uniqueness of solutions, $G(r, \theta)$ is independent of θ , so $k_g(r_0, \theta) = \left| \frac{G_r}{G} \right|$ is constant along the circle $r = r_0$.

(C) Geodesic curvature of a parallel of latitude on the unit 2-sphere.

On S^2 (radius 1), in spherical coordinates (φ, θ) with $\varphi = \text{colatitude}$ ($\varphi = 0$ north pole),

$$ds^2 = d\varphi^2 + \sin^2 \varphi d\theta^2.$$

A parallel is the curve $\varphi = \varphi_0$ (constant), θ varying. Here $G(\varphi) = \sin \varphi$, and the same computation as above yields

$$k_g(\varphi_0) = \left| \frac{(\sin \varphi)'}{\sin \varphi} \right|_{\varphi=\varphi_0} = \left| \frac{\cos \varphi_0}{\sin \varphi_0} \right| = |\cot \varphi_0|.$$

If instead λ denotes the usual latitude ($\lambda = 0$ at the equator), then $\varphi = \frac{\pi}{2} - \lambda$, so

$$k_g(\lambda) = |\tan \lambda|.$$

(With orientation, the sign is $\pm \cot \varphi_0$ / $\pm \tan \lambda$ depending on direction.)

Theory (for Problem 3: geodesic curvature of geodesic circles and parallels)

Fix a point $P \in S$. For sufficiently small $r > 0$, use *geodesic polar coordinates* (r, θ) centered at P , so that the metric has the form

$$ds^2 = dr^2 + G(r, \theta)^2 d\theta^2,$$

with

$$G(0, \theta) = 0, \quad G_r(0, \theta) = 1.$$

The geodesic circle of radius r_0 centered at P is the curve $\{r = r_0\}$.

(1) **Geodesic curvature of the circle $r = r_0$.** Along $r = r_0$, the coordinate vector ∂_θ has length $\|\partial_\theta\| = \sqrt{g_{\theta\theta}} = G(r_0, \theta)$. Hence a unit tangent field is

$$T = \frac{1}{G(r_0, \theta)} \partial_\theta.$$

For the metric $dr^2 + G^2 d\theta^2$, the relevant Christoffel symbol is

$$\Gamma_{\theta\theta}^r = -G G_r.$$

If s denotes arclength along the circle, then $ds = G(r_0, \theta) d\theta$, so that $d\theta/ds = 1/G(r_0, \theta)$. Therefore the radial component of $\nabla_T T$ is

$$(\nabla_T T)^r = \Gamma_{\theta\theta}^r \left(\frac{d\theta}{ds} \right)^2 = (-GG_r) \left(\frac{1}{G^2} \right) = -\frac{G_r}{G}.$$

Consequently, the geodesic curvature of the circle $r = r_0$ is

$$k_g(r_0, \theta) = \|\nabla_T T\| = \left| \frac{G_r(r_0, \theta)}{G(r_0, \theta)} \right|.$$

In particular, k_g is constant along the circle if $G(r_0, \theta)$ (equivalently, G_r/G) is independent of θ .

(2) Jacobi equation for G . For each fixed θ , the function $r \mapsto G(r, \theta)$ satisfies the scalar Jacobi equation

$$G_{rr}(r, \theta) + K(r, \theta) G(r, \theta) = 0,$$

where $K(r, \theta)$ is the Gaussian curvature at the point $\exp_P(r e_\theta)$. This follows from the Jacobi field obtained by varying the direction θ of the radial geodesics.

(3) Consequences for constancy of k_g .

- If $K \equiv K_0$ is constant in a neighborhood, then the Jacobi equation becomes

$$G_{rr} + K_0 G = 0, \quad G(0, \theta) = 0, \quad G_r(0, \theta) = 1,$$

whose solution is independent of θ by uniqueness. Hence $G(r, \theta) = G(r)$, and

$$k_g(r_0, \theta) = \left| \frac{G'(r_0)}{G(r_0)} \right|$$

is constant along each geodesic circle.

- More generally, if locally around P the curvature is constant along each geodesic circle, i.e. $K(r, \theta) = K(r)$, then the same uniqueness argument gives $G(r, \theta) = G(r)$, and again k_g is constant along the circle $r = r_0$.

(4) Parallels on the unit sphere. On the unit 2-sphere S^2 , in spherical coordinates (φ, θ) with colatitude φ , the metric is

$$ds^2 = d\varphi^2 + \sin^2 \varphi d\theta^2.$$

A parallel is given by $\varphi = \varphi_0$ (constant). Here $G(\varphi) = \sin \varphi$, hence

$$k_g(\varphi_0) = \left| \frac{(\sin \varphi)'}{\sin \varphi} \right|_{\varphi=\varphi_0} = \left| \frac{\cos \varphi_0}{\sin \varphi_0} \right| = |\cot \varphi_0|.$$

Equivalently, in terms of latitude $\lambda = \frac{\pi}{2} - \varphi$, one has $k_g(\lambda) = |\tan \lambda|$.

Geodesic polar coordinates and the geodesic circle. Fix a point $P \in S$. For sufficiently small $r > 0$, the exponential map $\exp_P: T_P S \rightarrow S$ is a diffeomorphism from a disk about $0 \in T_P S$ onto a neighborhood of P . Choose an oriented orthonormal basis in $T_P S$, and for each $\theta \in [0, 2\pi)$ let $e_\theta \in T_P S$ be the unit vector making angle θ . Then *geodesic polar coordinates* (r, θ) around P are defined by

$$(r, \theta) \longmapsto \exp_P(r e_\theta).$$

In these coordinates:

- r is the *geodesic distance* from P , i.e. $r = d(P, \exp_P(r e_\theta))$;
- θ records the *direction* in the tangent plane $T_P S$.

In geodesic polar coordinates the metric has the form

$$ds^2 = dr^2 + G(r, \theta)^2 d\theta^2,$$

where the initial conditions

$$G(0, \theta) = 0, \quad G_r(0, \theta) = 1$$

express that near P the metric looks Euclidean to first order (compare $dr^2 + r^2 d\theta^2$ in $\mathbb{R}^2$).

The *geodesic circle* of radius r_0 centered at P is the set of points at fixed geodesic distance r_0 from P ,

$$\{q \in S : d(P, q) = r_0\}.$$

In geodesic polar coordinates this set is exactly the coordinate curve

$$\{r = r_0\}.$$

It is called a *geodesic circle* because its radius is measured by geodesic distance (and it is built from radial geodesics $\theta = \text{const}$), not because the circle itself is a geodesic. In general, the curves $r = \text{const}$ are *not* geodesics (they typically have nonzero geodesic curvature), whereas the radial curves $\theta = \text{const}$ are geodesics by construction.

Geodesic polar coordinates on the unit sphere S^2 . Let $S^2 = \{x \in \mathbb{R}^3 : \|x\| = 1\}$ be the unit sphere, and fix a point $p \in S^2$.

1) Tangent space vs. tangent plane at p . The *tangent space* (a linear subspace of $\mathbb{R}^3$) is

$$T_p S^2 = \{v \in \mathbb{R}^3 : v \cdot p = 0\},$$

so it is a plane *through the origin* orthogonal to the radius vector p . The *tangent plane* (an affine plane through the point p) is

$$p + T_p S^2 = \{x \in \mathbb{R}^3 : p \cdot x = 1\},$$

which does *not* pass through the origin.

Choose an orthonormal basis (e_1, e_2) of $T_p S^2$. Any unit tangent direction is

$$e_\theta = \cos \theta e_1 + \sin \theta e_2, \quad \theta \in [0, 2\pi).$$

2) Radial geodesics from p . On S^2 , geodesics are great circles. The geodesic starting at p with initial velocity $e_\theta \in T_p S^2$ is

$$\gamma_\theta(r) = \exp_p(r e_\theta) = \cos r \, p + \sin r \, e_\theta,$$

where r is arc length along the geodesic (on the unit sphere, arc length equals the central angle in radians). Thus:

$$r = \text{geodesic distance from } p, \quad \theta = \text{direction in } T_p S^2.$$

3) The geodesic circle of radius r_0 . For $r_0 > 0$, the *geodesic circle* of radius r_0 centered at p is

$$C_{r_0}(p) = \{ q \in S^2 : d(p, q) = r_0 \}.$$

In geodesic polar coordinates (r, θ) centered at p , this circle is the coordinate curve

$$r = r_0, \quad \theta \in [0, 2\pi).$$

Using the explicit exponential-map formula,

$$C_{r_0}(p) = \{ \cos r_0 \, p + \sin r_0 \, e_\theta : \theta \in [0, 2\pi) \}.$$

Geometrically, $C_{r_0}(p)$ is a *small circle*: it is the intersection of S^2 with the plane

$$p \cdot x = \cos r_0.$$

If $r_0 = \frac{\pi}{2}$, then $p \cdot x = 0$ and the circle is a great circle (the “equator” relative to p). For $r_0 \neq \frac{\pi}{2}$, it is not a geodesic (its geodesic curvature is nonzero).

Problem 4 (Isometries are determined by a point and differential)

Problem. Let S be a connected surface and $f, g : S \rightarrow S$ two isometries. Assume that there exists $p \in S$ such that $f(p) = g(p)$ and $Df_p = Dg_p$. Show that $f(q) = g(q)$ for all $q \in S$.

Theory.

- An isometry sends geodesics to geodesics.
- A geodesic is uniquely determined by initial data $(\gamma(0), \gamma'(0))$ (ODE uniqueness).
- In a normal neighbourhood of p , points are of the form $\exp_p(v)$ and radial geodesics are unique.

Solution. Define $h := g^{-1} \circ f$. Then h is an isometry and

$$h(p) = g^{-1}(f(p)) = g^{-1}(g(p)) = p, \quad Dh_p = D(g^{-1})_{g(p)} \circ Df_p = (Dg_p)^{-1} \circ Df_p = \text{Id}.$$

Let $v \in T_p S$ and let γ_v be the geodesic with $\gamma_v(0) = p$ and $\gamma'_v(0) = v$. Since h is an isometry, $h \circ \gamma_v$ is a geodesic, and

$$(h \circ \gamma_v)(0) = h(p) = p, \quad (h \circ \gamma_v)'(0) = Dh_p(v) = v.$$

Thus $h \circ \gamma_v$ and γ_v satisfy the same geodesic equation with the same initial data, so by uniqueness

$$h(\gamma_v(t)) = \gamma_v(t)$$

for all t for which $\gamma_v(t)$ stays in a normal neighbourhood of p . Hence $h = \text{Id}$ on some open neighbourhood U of p .

Now let $F := \{x \in S : h(x) = x\}$. The set F is closed by continuity. It is open because if $x \in F$, then the above geodesic argument applied at x shows $h = \text{Id}$ on a neighbourhood of x . Since F is nonempty (it contains p), open and closed, and S is connected, we conclude $F = S$. Therefore $h = \text{Id}$ on S , i.e. $g^{-1} \circ f = \text{Id}$, and so $f = g$ everywhere.

Example. On $\mathbb{R}^2$, any isometry is $x \mapsto Ax + b$ with $A \in O(2)$. If $f(p) = g(p)$ and $Df_p = Dg_p$, then A and b coincide, hence $f = g$ globally.

Theory for Problem 4 (Rigidity of isometries from 1-jet data)

Let $(S, \langle \cdot, \cdot \rangle)$ be a connected Riemannian surface and let $f : S \rightarrow S$ be an isometry.

(1) Metric preservation and orthogonality of the differential. By definition of isometry,

$$\langle Df_p(X), Df_p(Y) \rangle_{f(p)} = \langle X, Y \rangle_p \quad \text{for all } p \in S, X, Y \in T_p S.$$

Hence $Df_p : T_p S \rightarrow T_{f(p)} S$ is an orthogonal linear map (it preserves lengths and angles).

(2) Isometries preserve the Levi-Civita connection. If ∇ denotes the Levi-Civita connection on S , then for all vector fields X, Y ,

$$Df(\nabla_X Y) = \nabla_{Df(X)} Df(Y).$$

(Equivalently, f is an affine map for ∇ .) This follows from the uniqueness of the Levi-Civita connection as the unique torsion-free, metric-compatible connection.

(3) Isometries send geodesics to geodesics and preserve speed. A curve γ is a geodesic iff $\nabla_{\dot{\gamma}} \dot{\gamma} = 0$. Using (2),

$$\nabla_{(f \circ \gamma)'} (f \circ \gamma)' = Df(\nabla_{\dot{\gamma}} \dot{\gamma}) = 0,$$

so $f \circ \gamma$ is geodesic whenever γ is. Moreover, since Df preserves norms,

$$\|(f \circ \gamma)'(t)\| = \|Df_{\gamma(t)}(\gamma'(t))\| = \|\gamma'(t)\|,$$

so in particular unit-speed geodesics are mapped to unit-speed geodesics.

(4) Uniqueness of geodesics from initial conditions. In local coordinates, geodesics satisfy the second-order ODE

$$x^{k''} + \Gamma_{ij}^k(x) x^{i'} x^{j'} = 0.$$

Therefore, by standard ODE uniqueness, if two geodesics γ, η satisfy $\gamma(0) = \eta(0)$ and $\gamma'(0) = \eta'(0)$, then $\gamma = \eta$ on their common interval of existence.

(5) Exponential map and normal neighbourhoods (optional viewpoint). For $p \in S$ define $\exp_p(v) = \gamma_v(1)$, where γ_v is the geodesic with $\gamma_v(0) = p$ and $\gamma'_v(0) = v$. For

$\|v\|$ small, $\exp_p$ is a diffeomorphism onto a normal neighbourhood U of p , and each $q \in U$ lies on a unique geodesic segment issuing from p .

(6) Consequence (idea used in the proof of Problem 4). If two isometries agree at p and have the same differential at p , then they send every geodesic starting at p with initial velocity v to geodesics with the same initial data. By (4) they coincide along these geodesics, hence they coincide on a neighbourhood of p , and then (by connectedness) everywhere on S .

Remark (“Rigidity” and “1-jet data” in Problem 4)

Rigidity (in this context). By *rigidity* we mean that an isometry has no freedom once enough local data are fixed: if two isometries $f, g : S \rightarrow S$ agree at one point p and have the same differential at p , then they must coincide everywhere on the connected surface S . Equivalently, an isometry is completely determined by its first-order behaviour at a single point.

1-jet data. For a smooth map $f : S \rightarrow S$, the *1-jet of f at p* is the pair

$$j_p^1 f := (f(p), Df_p),$$

consisting of the *value* of f at p (zeroth-order data) and the *differential* of f at p (first-order data). Thus

$$j_p^1 f = j_p^1 g \iff f(p) = g(p) \text{ and } Df_p = Dg_p.$$

Problem 4 states precisely that *isometries are 1-jet determined*: on a connected surface, equality of 1-jet data at one point forces $f = g$ on all of S .

Problem 5 (Geodesics locally minimize length)

Problem. (Geodesics are local minimizers of length.) Let p be a point on a surface S . Show that there exists an open set V containing p such that if $\gamma : [0, 1] \rightarrow V$ is a geodesic with $\gamma(0) = p$ and $\gamma(1) = q$ and $\alpha : [0, 1] \rightarrow S$ is a regular curve joining p to q , then

$$\ell(\gamma) \leq \ell(\alpha)$$

with equality if and only if α is a monotonic reparametrization of γ .

Theory.

- For any curve α from p to q , the intrinsic distance satisfies $d(p, q) \leq \ell(\alpha)$.
- For q in a sufficiently small (strongly convex) normal neighbourhood V of p , there is a unique minimizing geodesic segment $\gamma \subset V$ from p to q and $\ell(\gamma) = d(p, q)$.

Solution. Choose V to be a strongly convex normal neighbourhood of p . Then for every $q \in V$ there exists a unique minimizing geodesic segment $\gamma \subset V$ joining p to q , and it realizes the distance:

$$\ell(\gamma) = d(p, q).$$

Let α be any regular curve joining p to q (it may leave V). By definition of distance,

$$d(p, q) \leq \ell(\alpha).$$

Therefore

$$\ell(\gamma) = d(p, q) \leq \ell(\alpha),$$

which proves the inequality.

For equality: if $\ell(\alpha) = \ell(\gamma) = d(p, q)$, then α is a minimizing curve from p to q . In a strongly convex neighbourhood V , the minimizing curve from p to $q \in V$ is unique as an image, namely γ . Hence α must trace the same geodesic segment as γ . The only freedom is the parameter, so α is a monotone reparametrization of γ . Conversely, any monotone reparametrization preserves length, so equality holds.

Examples.

- On $\mathbb{R}^2$, the straight line segment from p to q minimizes length, and any detour is longer.
- On the unit sphere S^2 , if q is sufficiently close to p (distance $< \pi$), the shorter great-circle arc from p to q is minimizing inside a small normal neighbourhood.

Theory for Problem 5 (Geodesics locally minimize length)

(1) Length of a curve. If $\alpha : [0, 1] \rightarrow S$ is a piecewise C^1 regular curve on a Riemannian surface $(S, \langle \cdot, \cdot \rangle)$, its length is

$$\ell(\alpha) = \int_0^1 \|\alpha'(t)\| dt.$$

(2) Intrinsic distance. The Riemannian distance between $p, q \in S$ is defined by

$$d(p, q) = \inf \{ \ell(\alpha) : \alpha \text{ piecewise } C^1, \alpha(0) = p, \alpha(1) = q \}.$$

Consequently, for every curve α joining p to q one has the fundamental inequality

$$d(p, q) \leq \ell(\alpha).$$

(3) Normal neighbourhoods and the exponential map. For $p \in S$, the exponential map $\exp_p : T_p S \rightarrow S$ is defined by $\exp_p(v) = \gamma_v(1)$, where γ_v is the geodesic with $\gamma_v(0) = p$ and $\gamma'_v(0) = v$. There exists $r > 0$ such that $\exp_p$ restricts to a diffeomorphism from a ball $B_r(0) \subset T_p S$ onto an open set $V \subset S$ containing p (a *normal neighbourhood*). In such a V , each $q \in V$ is joined to p by a unique geodesic segment contained in V .

(4) Strongly convex neighbourhoods (local minimality and uniqueness). Shrinking V if necessary, one may assume V is *strongly convex*: for any $q \in V$ the unique geodesic segment $\gamma \subset V$ joining p to q is length-minimizing, and

$$\ell(\gamma) = d(p, q).$$

Moreover, in a strongly convex neighbourhood this minimizing geodesic segment is unique (as an image).

(5) Equality and monotone reparametrization. A curve $\alpha : [0, 1] \rightarrow S$ is a *monotone reparametrization* of a curve γ if

$$\alpha(t) = \gamma(\sigma(t))$$

for some C^1 function $\sigma : [0, 1] \rightarrow [0, 1]$ with $\sigma(0) = 0$, $\sigma(1) = 1$, and $\sigma'(t) \geq 0$. Such a reparametrization preserves length: $\ell(\alpha) = \ell(\gamma)$. In a strongly convex neighbourhood, equality $\ell(\alpha) = d(p, q)$ forces α to trace the unique minimizing geodesic segment, hence α must be a monotone reparametrization of γ .

(6) How the proof uses this. For $q \in V$ and the geodesic segment $\gamma \subset V$ from p to q ,

$$\ell(\gamma) = d(p, q) \leq \ell(\alpha)$$

for every curve α from p to q , with equality exactly for monotone reparametrizations of γ .

Remark (Geodesics are *not* defined as minimizers)

Geodesic definition (local straightness). A smooth curve γ on a Riemannian surface $(S, \langle \cdot, \cdot \rangle)$ is a *geodesic* if

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0,$$

i.e. its covariant acceleration vanishes. Equivalently, in local coordinates $x = (x^1, x^2)$ it satisfies the geodesic ODE

$$x^{k''} + \Gamma_{ij}^k(x) x^{i'} x^{j'} = 0.$$

This is the standard *definition*; it is not formulated in terms of length minimization.

Minimizing is a theorem (and only local in general). One proves that:

- If a regular curve minimizes length between its endpoints (under suitable smoothness assumptions), then it is a geodesic (it satisfies the Euler–Lagrange equation for the length/energy functional).
- Conversely, every geodesic segment is *locally* length-minimizing: for each point p there exists a neighbourhood V such that the unique geodesic segment in V joining p to $q \in V$ realizes the distance, $\ell(\gamma) = d(p, q)$ (this is the content of Problem 5).

Geodesics need not be globally minimizing. On the unit sphere S^2 , great circles are geodesics. Given two non-antipodal points, the shorter great-circle arc between them is minimizing, while the longer arc is still a geodesic but is *not* length-minimizing. More generally, a geodesic ceases to minimize length after passing a cut point.

Covariant vs. contravariant; covariant acceleration (geometric meaning)

Let (S, g) be a Riemannian surface and let $\gamma : [a, b] \rightarrow S$ be a smooth curve.

1) Why ordinary acceleration is not intrinsic. If $S \subset \mathbb{R}^3$ is an embedded surface, then γ can be viewed as a space curve and has the usual (Euclidean) acceleration $\gamma''(t) \in \mathbb{R}^3$. But $\gamma''(t)$ contains information both about how γ bends *within* the surface and how the surface itself sits in $\mathbb{R}^3$. At a point $p = \gamma(t)$ we can decompose

$$\gamma''(t) = (\gamma''(t))^\top + (\gamma''(t))^\perp,$$

where $(\cdot)^\top \in T_p S$ is the tangential projection and $(\cdot)^\perp$ is the normal projection.

Covariant acceleration. The intrinsic (surface) acceleration is the *covariant acceleration*

$$\nabla_{\dot{\gamma}} \dot{\gamma},$$

where ∇ is the Levi-Civita connection of g . For an embedded surface one has the concrete identity

$$\nabla_{\dot{\gamma}} \dot{\gamma} = (\gamma''(t))^\top,$$

i.e. “differentiate the velocity and keep only the tangential part”.

Geodesic condition. A curve γ is a geodesic iff

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0.$$

For an embedding $S \subset \mathbb{R}^3$ this means precisely that $\gamma''(t)$ is *purely normal*:

$$(\gamma'')^\top = 0 \iff \gamma''(t) \perp T_{\gamma(t)} S.$$

So a geodesic can have $\gamma'' \neq 0$ in $\mathbb{R}^3$ (e.g. great circles on a sphere), but it has no *intrinsic turning* inside the surface.

Covariant vs. contravariant components (index notation)

Choose local coordinates (u^1, u^2) on S , and write $\partial_i := \frac{\partial}{\partial u^i}$. A tangent vector $V \in T_p S$ is expressed as

$$V = V^i \partial_i.$$

The coefficients V^i (upper indices) are the **contravariant components**. Under a coordinate change $u^{i'} = u^{i'}(u^1, u^2)$ they transform like

$$V^{i'} = \frac{\partial u^{i'}}{\partial u^j} V^j.$$

The metric tensor has components

$$g_{ij} = \langle \partial_i, \partial_j \rangle.$$

Using g_{ij} we can *lower an index*:

$$V_i := g_{ij} V^j.$$

The numbers V_i (lower indices) are the **covariant components**; they transform with the inverse Jacobian:

$$V_{i'} = \frac{\partial u^j}{\partial u^{i'}} V_j.$$

(Conceptually: contravariant components describe vectors; covariant components describe covectors/linear functionals. The metric converts between them.)

Covariant derivative along a curve and the geodesic ODE

Let $V(t) = V^k(t) \partial_k$ be a vector field along $\gamma(t)$, and write $\dot{\gamma}(t) = u^i(t) \partial_i$. Because the basis ∂_k changes from point to point, the naive derivative $\frac{dV^k}{dt}$ is not intrinsic. The **covariant derivative along γ** is

$$\frac{DV^k}{dt} = \frac{dV^k}{dt} + \Gamma_{ij}^k(\gamma(t)) u^i(t) V^j(t), \quad \text{so} \quad \nabla_{\dot{\gamma}} V = \frac{DV}{dt}.$$

Taking $V = \dot{\gamma}$ gives the **geodesic equation** in coordinates:

$$\nabla_{\dot{\gamma}} \dot{\gamma} = 0 \iff u''^k + \Gamma_{ij}^k(u) u^i u^j = 0.$$

Diagram: decomposition of ambient acceleration and covariant acceleration

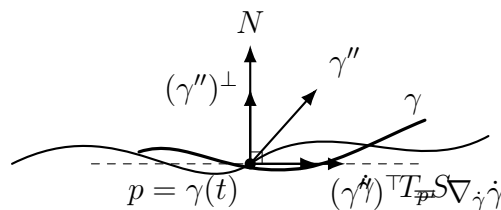


Figure 1: For an embedded surface $S \subset \mathbb{R}^3$, the ambient acceleration γ'' decomposes into a tangential part $(\gamma'')^\top$ and a normal part $(\gamma'')^\perp$. The intrinsic (covariant) acceleration is $\nabla_{\dot{\gamma}}\dot{\gamma} = (\gamma'')^\top$. A geodesic satisfies $(\gamma'')^\top = 0$, i.e. γ'' is purely normal.

Description (Figure ??). The figure illustrates the relation between the *ambient* (Euclidean) acceleration $\gamma''(t) \in \mathbb{R}^3$ of a curve γ on an embedded surface $S \subset \mathbb{R}^3$ and the *intrinsic* (surface) acceleration. At the point $p = \gamma(t)$ we decompose the vector $\gamma''(t)$ into its tangential and normal parts,

$$\gamma''(t) = (\gamma''(t))^\top + (\gamma''(t))^\perp, \quad (\gamma''(t))^\top \in T_p S, \quad (\gamma''(t))^\perp \perp T_p S.$$

The tangential component is precisely the *covariant acceleration*

$$\nabla_{\dot{\gamma}}\dot{\gamma} = (\gamma''(t))^\top,$$

while the normal component $(\gamma''(t))^\perp$ measures how the curve bends due to the embedding of S in $\mathbb{R}^3$. In particular, the geodesic condition $\nabla_{\dot{\gamma}}\dot{\gamma} = 0$ is equivalent to

$$(\gamma''(t))^\top = 0,$$

meaning that a geodesic has no *intrinsic* turning: any ambient acceleration is purely normal.

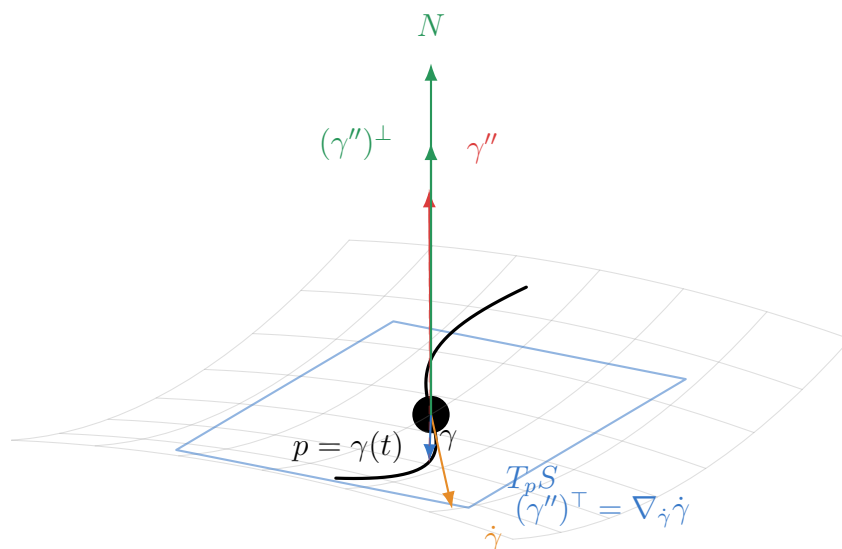


Figure 2: Pseudo-3D sketch: $\gamma'' = (\gamma'')^\top + (\gamma'')^\perp$, with $(\gamma'')^\top = \nabla_{\dot{\gamma}}\dot{\gamma}$. A geodesic satisfies $(\gamma'')^\top = 0$.

Problem 6 (Geodesic coordinates and the condition $G \equiv 1$)

Problem. Let P be a point on an embedded surface $S \subset \mathbb{R}^3$; consider the orthogonal parametrization $\phi : (-\varepsilon, \varepsilon)^2 \rightarrow V \subset S$ of a neighbourhood of P as constructed in lectures, where the curve $\phi(0, v)$ is a geodesic of unit speed, and for any $v_0 \in (-\varepsilon, \varepsilon)$ the curve $\phi(u, v_0)$ is a geodesic of unit speed. We showed that the first fundamental form was then

$$du^2 + G(u, v) dv^2$$

for some smooth function G . Prove that $G(u, v) = 1$ for all u, v if and only if the curves $\phi(u_0, v)$ are geodesics for all $u_0 \in (-\varepsilon, \varepsilon)$.

Theory. Here $E = 1$, $F = 0$, $G = G(u, v)$, so one computes (for example from the standard Christoffel formula) that

$$\Gamma_{vv}^u = -\frac{1}{2} \partial_u G.$$

Also, since $v \mapsto \phi(0, v)$ is unit-speed with parameter v , we have $G(0, v) = 1$.

Solution. ($\Rightarrow$) If $G \equiv 1$, then the metric is $ds^2 = du^2 + dv^2$, so all Christoffel symbols vanish in these coordinates. Hence the coordinate lines $u = \text{const}$ and $v = \text{const}$ satisfy the geodesic equations. In particular, each curve $v \mapsto \phi(u_0, v)$ is a geodesic.

($\Leftarrow$) Assume that for every u_0 the curve $v \mapsto \phi(u_0, v)$ is a geodesic. In coordinates this is $c(v) = (u_0, v)$, so $u(v) \equiv u_0$, hence $u' = 0$ and $u'' = 0$. The u -component of the geodesic equation becomes

$$0 = u'' + \Gamma_{vv}^u (v')^2 = \Gamma_{vv}^u (v')^2.$$

Since the curve is regular, $v' \neq 0$, and thus $\Gamma_{vv}^u = 0$ along c , i.e.

$$-\frac{1}{2} \partial_u G(u_0, v) = 0 \quad \text{for all } v.$$

Because this holds for all u_0 , we get $\partial_u G \equiv 0$ on the whole patch, hence $G(u, v) = H(v)$ depends only on v . Finally, the unit-speed assumption for $v \mapsto \phi(0, v)$ gives

$$1 = \|\partial_v \phi(0, v)\|^2 = G(0, v) = H(v),$$

so $H(v) \equiv 1$ and therefore $G(u, v) \equiv 1$.

Examples.

- Plane: $\phi(u, v) = (u, v)$ gives $ds^2 = du^2 + dv^2$ so $G \equiv 1$ and both coordinate families are geodesics.
- Cylinder of radius R : $\phi(u, v) = (R \cos(v/R), R \sin(v/R), u)$ yields $ds^2 = du^2 + dv^2$ so $G \equiv 1$, and the curves $u = \text{const}$ (circles around the cylinder) are geodesics.

Theory for Problem 6 (Orthogonal parametrization, Christoffel symbols, and an example)

Orthogonal parametrization. For a parametrization $\phi(u, v)$ of a surface, the first fundamental form is

$$ds^2 = E du^2 + 2F du dv + G dv^2, \quad E = \langle \phi_u, \phi_u \rangle, \quad F = \langle \phi_u, \phi_v \rangle, \quad G = \langle \phi_v, \phi_v \rangle.$$

The coordinate curves are orthogonal iff

$$\phi_u \perp \phi_v \iff F \equiv 0.$$

Geodesic (Fermi) coordinates along a geodesic. In the lecture construction one starts from a unit-speed geodesic $c(v) = \phi(0, v)$ and shoots unit-speed geodesics orthogonal to it. Concretely one may take a unit vector field $n(v) \in T_{c(v)}S$ orthogonal to $c'(v)$ (typically parallel transported along c) and define

$$\phi(u, v) = \exp_{c(v)}(u n(v)).$$

Then for each fixed v_0 , the curve $u \mapsto \phi(u, v_0)$ is a unit-speed geodesic, hence

$$E(u, v) = \|\phi_u(u, v)\|^2 \equiv 1.$$

Moreover the construction is orthogonal, so $F \equiv 0$. Therefore the metric has the special form

$$ds^2 = du^2 + G(u, v) dv^2.$$

Since $v \mapsto \phi(0, v)$ is assumed unit-speed with parameter v , we also have

$$G(0, v) = \|\phi_v(0, v)\|^2 = 1.$$

Christoffel symbols for $ds^2 = du^2 + G dv^2$. Here $g_{11} = 1$, $g_{12} = 0$, $g_{22} = G$, and $g^{11} = 1$, $g^{22} = 1/G$. The nonzero Christoffel symbols are

$$\Gamma_{vv}^u = -\frac{1}{2} \partial_u G, \quad \Gamma_{uv}^v = \Gamma_{vu}^v = \frac{1}{2G} \partial_u G, \quad \Gamma_{vv}^v = \frac{1}{2G} \partial_v G,$$

and all others vanish.

Geodesic criterion for the curves $u = \text{const}$. Consider the coordinate curve $v \mapsto (u_0, v)$, so $u(v) \equiv u_0$ and hence $u' = 0$, $u'' = 0$. The u -component of the geodesic equation is

$$u'' + \Gamma_{vv}^u (v')^2 = 0.$$

For a regular parametrization ($v' \neq 0$), this implies $\Gamma_{vv}^u = 0$, hence

$$-\frac{1}{2} \partial_u G(u_0, v) = 0 \iff \partial_u G(u_0, v) = 0.$$

If this holds for all u_0 , then $\partial_u G \equiv 0$ and thus $G(u, v) = H(v)$ depends only on v . Using $G(0, v) = 1$ gives $H(v) \equiv 1$, hence $G \equiv 1$. Conversely, if $G \equiv 1$, then $ds^2 = du^2 + dv^2$ and all Christoffel symbols vanish, so the curves $u = \text{const}$ are geodesics.

Example (unit sphere). On the unit sphere S^2 , take

$$\phi(u, v) = (\cos u \cos v, \cos u \sin v, \sin u).$$

Then

$$ds^2 = du^2 + \cos^2(u) dv^2,$$

so $E = 1$, $F = 0$, $G(u, v) = \cos^2 u$ and $G(0, v) = 1$. The curves $v = \text{const}$ (meridians) are unit-speed geodesics in parameter u , while the curves $u = \text{const}$ (parallels) are not geodesics unless $u = 0$, in accordance with the criterion $\partial_u G = 0 \iff G \equiv 1$.

Christoffel symbols from the Levi–Civita connection (and application to $ds^2 = du^2 + G dv^2$)

1) Levi–Civita (Koszul) formula in coordinates. Let (S, g) be a Riemannian surface and let (x^1, x^2) be local coordinates with $\partial_i := \frac{\partial}{\partial x^i}$. The Levi–Civita connection ∇ is characterized by torsion-freeness and metric-compatibility. In coordinates, since $[\partial_i, \partial_j] = 0$, the Koszul formula reduces to

$$2\langle \nabla_{\partial_i} \partial_j, \partial_k \rangle = \partial_i \langle \partial_j, \partial_k \rangle + \partial_j \langle \partial_i, \partial_k \rangle - \partial_k \langle \partial_i, \partial_j \rangle.$$

Writing $g_{ij} = \langle \partial_i, \partial_j \rangle$, this becomes

$$2\langle \nabla_{\partial_i} \partial_j, \partial_k \rangle = \partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}.$$

Expanding $\nabla_{\partial_i} \partial_j$ in the coordinate basis,

$$\nabla_{\partial_i} \partial_j = \Gamma_{ij}^m \partial_m,$$

we obtain

$$\langle \nabla_{\partial_i} \partial_j, \partial_k \rangle = \Gamma_{ij}^m g_{mk}.$$

Hence

$$2\Gamma_{ij}^m g_{mk} = \partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}.$$

Multiplying by the inverse metric $g^{k\ell}$ (so that $g_{mk} g^{k\ell} = \delta_m^\ell$) yields the standard Christoffel symbol formula:

$$\Gamma_{ij}^\ell = \frac{1}{2} g^{\ell k} (\partial_i g_{jk} + \partial_j g_{ik} - \partial_k g_{ij}).$$

2) Application to the metric $ds^2 = du^2 + G(u, v) dv^2$. Let $(x^1, x^2) = (u, v)$. Then

$$(g_{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & G(u, v) \end{pmatrix}, \quad (g^{ij}) = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1}{G(u, v)} \end{pmatrix}.$$

Derivatives:

$$\partial_u g_{11} = 0, \quad \partial_v g_{11} = 0, \quad \partial_u g_{22} = \partial_u G, \quad \partial_v g_{22} = \partial_v G, \quad g_{12} = 0 \Rightarrow \partial_u g_{12} = \partial_v g_{12} = 0.$$

Using the boxed formula one finds the only potentially nonzero Christoffel symbols:

$$\Gamma_{vv}^u = -\frac{1}{2} \partial_u G, \quad \Gamma_{uv}^v = \Gamma_{vu}^v = \frac{1}{2G} \partial_u G, \quad \Gamma_{vv}^v = \frac{1}{2G} \partial_v G.$$

All other Γ_{ij}^k vanish for this metric.

3) When do the Christoffel symbols vanish? The Levi–Civita connection itself is not “zero” in general; however, in a given coordinate system it may happen that all Christoffel symbols Γ_{ij}^k vanish. In the present coordinates, if $G \equiv 1$ then

$$ds^2 = du^2 + dv^2,$$

and indeed all $\Gamma_{ij}^k = 0$ (the metric coefficients are constant). This is exactly the situation in which the geodesic equations reduce to $u'' = 0$, $v'' = 0$ in these coordinates.

4) Example (sphere in orthogonal coordinates). On the unit sphere S^2 , consider

$$\phi(u, v) = (\cos u \cos v, \cos u \sin v, \sin u),$$

for which

$$ds^2 = du^2 + \cos^2(u) dv^2, \quad \text{so} \quad G(u, v) = \cos^2 u.$$

Then $\partial_u G = -2 \sin u \cos u$ and $\partial_v G = 0$, hence

$$\Gamma_{vv}^u = -\frac{1}{2} \partial_u G = \sin u \cos u, \quad \Gamma_{uv}^v = \Gamma_{vu}^v = \frac{1}{2G} \partial_u G = -\tan u, \quad \Gamma_{vv}^v = 0.$$

These vanish only at $u = 0$, reflecting that the parallels $u = \text{const}$ are geodesics only for $u = 0$ (the equator), while meridians $v = \text{const}$ are geodesics for all u .

Why geodesic coordinates are used in Problem 6 (and the sphere chart interpretation)

Geodesic coordinates in Problem 6. In Problem 6 the parametrization

$$\phi : (-\varepsilon, \varepsilon)^2 \rightarrow S, \quad (u, v) \mapsto \phi(u, v),$$

is constructed so that the *coordinate curves are geodesics*. This choice simplifies both the metric and the geodesic equations, and reduces the geometry to understanding a single coefficient $G(u, v)$.

What “geodesic coordinates” means here. By assumption, for every fixed v_0 the curve

$$u \mapsto \phi(u, v_0)$$

is a *unit-speed geodesic*. Therefore

$$E(u, v) = \langle \phi_u, \phi_u \rangle \equiv 1.$$

Moreover the parametrization is *orthogonal*, so

$$F(u, v) = \langle \phi_u, \phi_v \rangle \equiv 0.$$

Hence the first fundamental form becomes

$$ds^2 = du^2 + G(u, v) dv^2, \quad \text{where} \quad G(u, v) = \langle \phi_v, \phi_v \rangle.$$

Thus the whole problem reduces to understanding the single function $G(u, v)$.

Relation to Fermi coordinates. This construction is a special case of *Fermi coordinates along a geodesic*: one starts from a “base” geodesic $v \mapsto \phi(0, v)$ and then fills a neighbourhood by geodesics orthogonal to it, namely $u \mapsto \phi(u, v_0)$.

Sphere example: the chart, the base curve, and the coordinate curves

Consider the unit sphere S^2 and the parametrization

$$\phi(u, v) = (\cos u \cos v, \cos u \sin v, \sin u) \in S^2.$$

What chart is this? This is the standard “latitude–longitude” chart:

- u is the *latitude* (angle from the equatorial plane), typically $u \in (-\frac{\pi}{2}, \frac{\pi}{2})$;
- v is the *longitude* (angle around the z -axis), typically $v \in (0, 2\pi)$.

Indeed, the third coordinate is

$$z = \sin u,$$

so u controls the height. For fixed u , the radius in the xy -plane is $\cos u$, producing circles of latitude.

Which curve is the “base geodesic”? In Problem 6 the construction starts from $v \mapsto \phi(0, v)$. Here

$$\phi(0, v) = (\cos v, \sin v, 0),$$

which is the *equator*. It is a great circle, hence a geodesic. Moreover it is unit-speed in the parameter v because

$$\|\partial_v \phi(0, v)\|^2 = G(0, v) = 1.$$

What are the u -curves $\phi(u, v_0)$? Fix $v = v_0$. Then

$$\phi(u, v_0) = (\cos u \cos v_0, \cos u \sin v_0, \sin u).$$

This curve lies in the plane spanned by the z -axis and the direction $(\cos v_0, \sin v_0, 0)$. That plane passes through the origin, so its intersection with S^2 is a *great circle*. Thus these curves are the *meridians*, hence geodesics. Also $\|\phi_u\| \equiv 1$, so u is arc-length along them.

What are the v -curves $\phi(u_0, v)$? Fix $u = u_0$. Then $z = \sin u_0$ is constant and

$$v \mapsto (\cos u_0 \cos v, \cos u_0 \sin v, \sin u_0)$$

is a *parallel* (circle of latitude). These are not geodesics unless $u_0 = 0$ (the equator).

Why this matters for Problem 6. For this chart one computes

$$ds^2 = du^2 + \cos^2(u) dv^2, \quad \text{so} \quad G(u, v) = \cos^2 u.$$

Hence $G \not\equiv 1$, and accordingly the curves $u = \text{const}$ (parallels) fail to be geodesics except at $u = 0$. This matches the statement of Problem 6:

$$G \equiv 1 \iff \text{for every } u_0, \text{ the curve } v \mapsto \phi(u_0, v) \text{ is a geodesic.}$$

Examples illustrating Problem 6

Example A (model case): the plane $\mathbb{R}^2$ with $G \equiv 1$

Take the standard parametrization

$$\phi(u, v) = (u, v) \in \mathbb{R}^2.$$

Then

$$\phi_u = (1, 0), \quad \phi_v = (0, 1),$$

so

$$E = \langle \phi_u, \phi_u \rangle = 1, \quad F = \langle \phi_u, \phi_v \rangle = 0, \quad G = \langle \phi_v, \phi_v \rangle = 1.$$

Hence the first fundamental form is

$$ds^2 = du^2 + dv^2, \quad \text{so in particular} \quad G \equiv 1.$$

The coordinate curves are straight lines, hence unit-speed geodesics:

- the base curve $\phi(0, v) = (0, v)$ is a straight line (unit-speed geodesic);
- for fixed v_0 , the u -curves $\phi(u, v_0) = (u, v_0)$ are straight lines (unit-speed geodesics);
- for fixed u_0 , the v -curves $\phi(u_0, v) = (u_0, v)$ are straight lines (unit-speed geodesics).

This represents the local “Euclidean” behaviour characterized in Problem 6.

Example B: cylinder (a genuine grid of geodesics) with $G \equiv 1$

Let S be the cylinder of radius $R > 0$:

$$S = \{(x, y, z) \in \mathbb{R}^3 : x^2 + y^2 = R^2\}.$$

Define

$$\phi(u, v) = \left(R \cos\left(\frac{v}{R}\right), R \sin\left(\frac{v}{R}\right), u \right).$$

The scaling v/R is chosen so that v is *arc-length* around each circular cross-section. Indeed,

$$\phi_u = (0, 0, 1) \Rightarrow \|\phi_u\| = 1 \Rightarrow E = 1,$$

and

$$\phi_v = \left(-\sin\left(\frac{v}{R}\right), \cos\left(\frac{v}{R}\right), 0 \right) \Rightarrow \|\phi_v\| = 1 \Rightarrow G = 1.$$

Moreover $\langle \phi_u, \phi_v \rangle = 0$, hence $F = 0$. Therefore

$$ds^2 = du^2 + dv^2, \quad \text{so} \quad G \equiv 1,$$

exactly as in Problem 6.

Coordinate curves.

- The base curve

$$\phi(0, v) = \left(R \cos\left(\frac{v}{R}\right), R \sin\left(\frac{v}{R}\right), 0 \right)$$

is a horizontal circle. On the cylinder it is a geodesic (its curvature vector points radially, hence normal to the surface), and it is unit-speed by the v/R scaling.

- For fixed v_0 , the u -curves $u \mapsto \phi(u, v_0)$ are vertical generators (straight lines in $\mathbb{R}^3$), hence unit-speed geodesics.
- For fixed u_0 , the v -curves $v \mapsto \phi(u_0, v)$ are horizontal circles, again unit-speed geodesics.

Thus the cylinder provides a genuine “geodesic grid” with $G \equiv 1$.

Example C (contrast): the sphere S^2 with $G(u, v) = \cos^2 u$

For completeness, consider

$$\phi(u, v) = (\cos u \cos v, \cos u \sin v, \sin u) \in S^2.$$

A direct computation gives

$$ds^2 = du^2 + \cos^2(u) dv^2, \quad \text{so} \quad G(u, v) = \cos^2 u \neq 1 \text{ unless } u = 0.$$

Here the base curve $\phi(0, v)$ is the equator (a unit-speed geodesic), and the u -curves (meridians) are geodesics. However, the v -curves (parallels) are not geodesics unless $u = 0$, in agreement with the criterion in Problem 6.

Problem 7 (Sign change of Gaussian curvature on a non-spherical compact surface)

Statement. Let S be a compact connected orientable surface in $\mathbb{R}^3$ which is not homeomorphic to a sphere. Prove that there are points on S where the Gaussian curvature is positive, negative, and zero.

Solution. *Step 1: $K > 0$ somewhere.* Fix a unit vector $v \in \mathbb{R}^3$ and consider the height function

$$f : S \rightarrow \mathbb{R}, \quad f(p) = \langle p, v \rangle.$$

Since S is compact, f attains a maximum at some $p_0 \in S$. At a maximum, the tangential gradient vanishes, so v is normal to S at p_0 . Moreover, the Hessian of f restricted to $T_{p_0}S$ is negative semidefinite. Up to sign (depending on the orientation of the unit normal), this Hessian equals the second fundamental form, so the principal curvatures satisfy $k_1(p_0) \geq 0$ and $k_2(p_0) \geq 0$, hence $K(p_0) = k_1(p_0)k_2(p_0) \geq 0$.

Choose v generically so that the maximum p_0 is *nondegenerate* (Morse), i.e. the Hessian is negative definite. Then $k_1(p_0) > 0$ and $k_2(p_0) > 0$, so

$$K(p_0) > 0.$$

Step 2: $K < 0$ somewhere. Since S is compact, connected, orientable, and not homeomorphic to S^2 , it has genus $g \geq 1$ and

$$\chi(S) = 2 - 2g \leq 0.$$

By Gauss–Bonnet (for a closed surface),

$$\int_S K \, dA = 2\pi \chi(S) \leq 0.$$

But $K > 0$ at p_0 , so K cannot be nonnegative everywhere unless the integral were > 0 or (in genus 1) could not balance to 0. More precisely:

- If $g \geq 2$, then $2\pi\chi(S) < 0$, so necessarily $K < 0$ somewhere.
- If $g = 1$, then $\int_S K \, dA = 0$, but since $K > 0$ somewhere, there must be points where $K < 0$ to cancel the positive contribution.

Thus $K < 0$ somewhere on S .

Step 3: $K = 0$ somewhere. The function $K : S \rightarrow \mathbb{R}$ is continuous and S is connected. Since K takes both positive and negative values, by the intermediate value theorem there exists $p \in S$ such that

$$K(p) = 0.$$

This proves that K is positive, negative, and zero at some points of S .

Examples.

- A standard torus of revolution in $\mathbb{R}^3$ has regions with $K > 0$ (outer side), $K < 0$ (inner neck), and $K = 0$ along two parabolic circles separating them.
- A smooth embedded genus-2 surface (“double torus”) similarly has elliptic regions ($K > 0$), saddle regions ($K < 0$), and parabolic curves ($K = 0$).

Problem 8 (Equal Gauss-image areas across a closed geodesic in $K > 0$)

Statement. Let S be a compact oriented surface with positive Gaussian curvature and let $N : S \rightarrow S^2$ be the Gauss map. Let γ be a simple closed geodesic in S , and let A and B be the regions which have γ as a common boundary. Show that $N(A)$ and $N(B)$ have the same area.

Solution. *Step 1: S is topologically a sphere.* Since $K > 0$ everywhere, Gauss–Bonnet gives

$$\int_S K \, dA = 2\pi \chi(S) > 0 \quad \Rightarrow \quad \chi(S) > 0.$$

For compact orientable surfaces, $\chi(S) > 0$ implies S is homeomorphic to S^2 .

Step 2: A and B are disks. On a sphere, any simple closed curve separates the surface into two regions each homeomorphic to a disk, hence

$$\chi(A) = \chi(B) = 1.$$

Step 3: Apply Gauss–Bonnet to A and B . For a compact region D with smooth boundary, Gauss–Bonnet says

$$\int_D K \, dA + \int_{\partial D} k_g \, ds = 2\pi \chi(D),$$

where k_g is the geodesic curvature of ∂D . Here $\partial A = \gamma = \partial B$, and γ is a geodesic, so $k_g \equiv 0$ on γ . Therefore,

$$\int_A K \, dA = 2\pi \chi(A) = 2\pi, \quad \int_B K \, dA = 2\pi \chi(B) = 2\pi,$$

so

$$\int_A K \, dA = \int_B K \, dA.$$

Step 4: Convert curvature integrals into areas on the Gauss sphere. The Jacobian of the Gauss map satisfies

$$|\det(dN)| = |K|.$$

Since $K > 0$, N is orientation-preserving locally and the area of the image is (with multiplicity)

$$\text{Area}(N(D)) = \int_D K \, dA.$$

Applying this to $D = A$ and $D = B$ yields

$$\text{Area}(N(A)) = \int_A K \, dA \quad \text{and} \quad \text{Area}(N(B)) = \int_B K \, dA,$$

hence $\text{Area}(N(A)) = \text{Area}(N(B))$.

Examples.

- If $S = S^2$ is the unit sphere, then N is the identity (up to orientation) and any great circle geodesic splits S^2 into two hemispheres of equal area 2π .
- If S is a smooth strictly convex surface (e.g. an ellipsoid), then $K > 0$ and N is a diffeomorphism onto S^2 . Any simple closed geodesic γ splits S into disks A, B whose Gauss images have equal area.

Problem 9 (No geodesic bigons in $K \leq 0$; uniqueness on a negatively curved cylinder)

Statement.

1. Let S be an orientable surface with Gaussian curvature $K \leq 0$. Show that two geodesics γ_1 and γ_2 which start from a point $p \in S$ will not meet again at a point q in such a way that the traces (images) of γ_1 and γ_2 form the boundary of a domain homeomorphic to a disk.
2. Let S be a surface homeomorphic to a cylinder and with negative Gaussian curvature. Show that S has at most one simple closed geodesic. Does the result remain true if “negative” is replaced with “non-positive”?

Solution.

(i) Assume, for contradiction, that γ_1, γ_2 start at p , meet again at q , and bound a domain $D \subset S$ homeomorphic to a disk. Then ∂D consists of two geodesic arcs, so along the smooth parts of ∂D we have $k_g \equiv 0$. The boundary has corners at p and q with interior angles $\theta_p, \theta_q \in (0, \pi]$.

Gauss–Bonnet for a disk with corners gives

$$\int_D K dA + \int_{\partial D} k_g ds + \sum_{x \in \{p, q\}} (\pi - \theta_x) = 2\pi.$$

Since $k_g \equiv 0$ on the geodesic arcs and $K \leq 0$, this yields

$$2\pi = \int_D K dA + (\pi - \theta_p) + (\pi - \theta_q) \leq 0 + 2\pi - (\theta_p + \theta_q).$$

Hence $\theta_p + \theta_q \leq 0$, impossible. Therefore such a disk-bounding “geodesic bigon” cannot exist.

(ii) Suppose S is homeomorphic to a cylinder and $K < 0$ everywhere. Assume there are two distinct simple closed geodesics γ_0 and γ_1 . Then they bound a compact region $A \subset S$ homeomorphic to an annulus. For an annulus, $\chi(A) = 0$, and since each boundary component is a geodesic, $k_g \equiv 0$ on ∂A . Gauss–Bonnet gives

$$\int_A K dA + \int_{\partial A} k_g ds = 2\pi\chi(A) = 0 \implies \int_A K dA = 0.$$

But $K < 0$ on A implies $\int_A K dA < 0$, a contradiction. Hence S has at most one simple closed geodesic.

Non-positive curvature? The conclusion need not hold for $K \leq 0$. Indeed, if $K \equiv 0$ on an annular region, the above Gauss–Bonnet equation allows $\int_A K dA = 0$ with no contradiction. For example, the standard round cylinder $S^1 \times \mathbb{R} \subset \mathbb{R}^3$ has $K \equiv 0$ and contains infinitely many simple closed geodesics (all circles at constant height).

Examples.

- For (i): in the Euclidean plane ($K = 0$) or hyperbolic plane ($K < 0$), two distinct geodesics from p cannot meet again and bound a disk (no geodesic bigons).

- For (ii): a hyperbolic cylinder (a quotient of $\mathbb{H}^2$ by a hyperbolic isometry) has a unique simple closed geodesic (the projection of the axis of the isometry).
- For the failure when $K \leq 0$: the round cylinder in $\mathbb{R}^3$ has $K \equiv 0$ and infinitely many simple closed geodesics.

Problem 10 (Holonomy angle and Gaussian curvature)

Statement. Let $\phi : U \rightarrow S$ be an orthogonal parametrization around a point p (so $F = 0$ on U). Let $\alpha : [0, \ell] \rightarrow \phi(U)$ be a smooth simple closed curve parametrized by arc-length, enclosing a domain R . Fix a unit vector $w_0 \in T_{\alpha(0)}S$ and let $W(t)$ be the parallel transport of w_0 along α . Let $\psi(t)$ be a differentiable determination of the angle from ϕ_u to $W(t)$. Show that

$$\psi(\ell) - \psi(0) = \int_R K \, dA.$$

Let S be a connected surface. Use the above to show that if the parallel transport between any two points does not depend on the curve joining the points, then the Gaussian curvature of S is zero.

Solution. *Setup:* an orthonormal frame and the connection 1-form. Since ϕ is orthogonal, write the first fundamental form as

$$ds^2 = E \, du^2 + G \, dv^2 \quad (F = 0),$$

and define an oriented orthonormal frame on $\phi(U)$ by

$$e_1 = \frac{\phi_u}{\sqrt{E}}, \quad e_2 = \frac{\phi_v}{\sqrt{G}}.$$

Let ω_{12} be the Levi-Civita connection 1-form of this frame, characterized by

$$\nabla e_1 = \omega_{12} e_2, \quad \nabla e_2 = -\omega_{12} e_1,$$

equivalently $\omega_{12}(X) = \langle \nabla_X e_1, e_2 \rangle$.

Step 1: relate the angle change to ω_{12} . Along $\alpha(t)$, write the transported unit vector $W(t)$ in the frame:

$$W(t) = \cos \psi(t) e_1(\alpha(t)) + \sin \psi(t) e_2(\alpha(t)).$$

Differentiate covariantly along α and use that W is parallel:

$$\nabla_{\alpha'} W = 0.$$

Using the frame relations for $\nabla e_1, \nabla e_2$ and collecting the coefficients of e_1, e_2 gives

$$\psi'(t) = \omega_{12}(\alpha'(t)).$$

(With the chosen orientation conventions, the sign is as above; reversing the orientation changes both sides consistently.)

Integrating from 0 to ℓ yields

$$\psi(\ell) - \psi(0) = \int_0^\ell \omega_{12}(\alpha'(t)) \, dt = \int_\alpha \omega_{12}.$$

Step 2: apply the structure equation and Stokes. For an oriented orthonormal frame on a surface, Cartan's second structure equation gives

$$d\omega_{12} = K dA,$$

where $dA = \sqrt{EG} du dv$ is the area form on S induced by the chosen orientation.

Since $\alpha = \partial R$ is positively oriented boundary of R , Stokes' theorem yields

$$\int_{\alpha} \omega_{12} = \int_{\partial R} \omega_{12} = \int_R d\omega_{12} = \int_R K dA.$$

Combining with Step 1 proves

$$\psi(\ell) - \psi(0) = \int_R K dA.$$

Consequence (path-independence $\Rightarrow K \equiv 0$). Assume parallel transport between any two points is independent of the chosen curve. Then for every smooth loop α based at p , the parallel transport around α is the identity on $T_p S$. In particular, for a simple closed loop $\alpha = \partial R$ contained in a coordinate patch, we may choose ψ continuously so that $\psi(\ell) = \psi(0)$ (no net rotation). By the formula just proved,

$$0 = \psi(\ell) - \psi(0) = \int_R K dA$$

for every sufficiently small region R .

Fix any $p \in S$ and take a family of regions R_ε shrinking to p . Then

$$0 = \frac{1}{\text{Area}(R_\varepsilon)} \int_{R_\varepsilon} K dA \longrightarrow K(p) \quad \text{as } \varepsilon \rightarrow 0,$$

by continuity of K (the average value of a continuous function over shrinking regions tends to its value at the point). Hence $K(p) = 0$ for all p , i.e. $K \equiv 0$ on S .

Examples.

- On the unit sphere S^2 , $K \equiv 1$, so the angle change after transporting around a region R equals $\text{Area}(R)$ on the sphere.
- On a plane (or any developable surface patch), $K \equiv 0$, and parallel transport is path-independent locally: the holonomy angle around any small loop is 0.

Theory (Problem 10): parallel transport, connection form, and curvature

Let S be an oriented surface with the induced Riemannian metric and Levi-Civita connection ∇ .

1) Oriented orthonormal frame and the connection 1-form. Let (e_1, e_2) be a local oriented orthonormal tangent frame on an open set $U \subset S$. There exists a unique 1-form ω_{12} on U (the connection 1-form) such that for every tangent vector field X

$$\nabla_X e_1 = \omega_{12}(X) e_2, \quad \nabla_X e_2 = -\omega_{12}(X) e_1.$$

Equivalently,

$$\omega_{12}(X) = \langle \nabla_X e_1, e_2 \rangle,$$

so ω_{12} measures the infinitesimal rotation of the frame.

2) Canonical orthonormal frame in orthogonal coordinates. If $\phi(u, v)$ is an orthogonal parametrization (so $F = 0$ and $ds^2 = E du^2 + G dv^2$), then

$$e_1 = \frac{\phi_u}{\sqrt{E}}, \quad e_2 = \frac{\phi_v}{\sqrt{G}}$$

is an oriented orthonormal frame on U .

3) Angle equation for a parallel vector field. Let $\alpha : [0, \ell] \rightarrow U$ be a smooth curve and let $W(t) \in T_{\alpha(t)}S$ be a unit vector field along α . Write

$$W(t) = \cos \psi(t) e_1(\alpha(t)) + \sin \psi(t) e_2(\alpha(t)),$$

where $\psi(t)$ is a differentiable determination of the angle from e_1 to W . If W is parallel along α , i.e. $\nabla_{\alpha'(t)} W(t) = 0$, then

$$\boxed{\psi'(t) = \omega_{12}(\alpha'(t))},$$

and hence

$$\boxed{\psi(\ell) - \psi(0) = \int_{\alpha} \omega_{12}}.$$

4) Curvature 2-form (Cartan structure equation in dimension 2). For an oriented orthonormal frame on a surface,

$$\boxed{d\omega_{12} = K dA},$$

where K is the Gaussian curvature and dA is the area form ($dA = \sqrt{EG - F^2} du dv$, and in particular $dA = \sqrt{EG} du dv$ when $F = 0$).

5) Stokes: holonomy equals curvature flux. If $\alpha = \partial R$ is the positively oriented boundary of a region $R \subset U$, then by Stokes' theorem,

$$\int_{\alpha} \omega_{12} = \int_{\partial R} \omega_{12} = \int_R d\omega_{12} = \int_R K dA.$$

Combining this with (3) gives the holonomy formula used in Problem 10:

$$\boxed{\psi(\ell) - \psi(0) = \int_R K dA}.$$

Theory for Problem 10: parallel transport, connection 1-form, and holonomy

Let S be an oriented surface with the induced Riemannian metric and Levi-Civita connection ∇ .

1) Oriented orthonormal frame and the connection 1-form. Let (e_1, e_2) be a local *oriented orthonormal tangent frame* on an open set $U \subset S$. There exists a unique 1-form ω_{12} on U (the *connection 1-form*) such that for every tangent vector field X

$$\nabla_X e_1 = \omega_{12}(X) e_2, \quad \nabla_X e_2 = -\omega_{12}(X) e_1.$$

Equivalently,

$$\omega_{12}(X) = \langle \nabla_X e_1, e_2 \rangle.$$

Intuitively, $\omega_{12}(X)$ measures the instantaneous angular velocity of the frame when moving in direction X .

2) Orthogonal coordinates and a canonical orthonormal frame. If $\phi(u, v)$ is an *orthogonal parametrization* on U (so $F = 0$ and $ds^2 = E du^2 + G dv^2$), a standard choice is

$$e_1 = \frac{\phi_u}{\sqrt{E}}, \quad e_2 = \frac{\phi_v}{\sqrt{G}},$$

which is an oriented orthonormal frame on U .

3) Angle equation for a parallel vector field. Let $\alpha : [0, \ell] \rightarrow U$ be a smooth curve and let $W(t) \in T_{\alpha(t)}S$ be a unit vector field along α . Write W in the frame (e_1, e_2) as

$$W(t) = \cos \psi(t) e_1(\alpha(t)) + \sin \psi(t) e_2(\alpha(t)),$$

where $\psi(t)$ is a differentiable determination of the angle from e_1 to W . If W is *parallel* along α , i.e.

$$\nabla_{\alpha'(t)} W(t) = 0,$$

then a direct computation using the identities in (1) gives

$$\boxed{\psi'(t) = \omega_{12}(\alpha'(t))}.$$

Hence the net rotation of W along α is

$$\boxed{\psi(\ell) - \psi(0) = \int_{\alpha} \omega_{12}}.$$

4) Curvature 2-form (Cartan structure equation in dimension 2). For an oriented orthonormal frame on a surface, the connection form satisfies

$$\boxed{d\omega_{12} = K dA},$$

where K is the Gaussian curvature and dA is the area form of the induced metric (in local coordinates (u, v) , $dA = \sqrt{EG - F^2} du dv$ and in particular $dA = \sqrt{EG} du dv$ when $F = 0$).

5) Stokes: holonomy equals curvature flux. If $\alpha = \partial R$ is the positively oriented boundary of a region $R \subset U$, then Stokes' theorem gives

$$\int_{\alpha} \omega_{12} = \int_{\partial R} \omega_{12} = \int_R d\omega_{12}.$$

Combining with $d\omega_{12} = K dA$ yields

$$\boxed{\int_{\alpha} \omega_{12} = \int_R K dA}.$$

Together with (3), this is exactly the holonomy formula used in Problem 10:

$$\psi(\ell) - \psi(0) = \int_R K dA.$$

Consequence (path-independence $\Rightarrow K \equiv 0$). If parallel transport between any two points is independent of the curve joining them, then parallel transport around every sufficiently small closed loop is the identity, so $\psi(\ell) - \psi(0) = 0$ for such loops. By the above formula, $\int_R K dA = 0$ for all small regions R , hence by continuity $K(p) = 0$ at every point $p \in S$, so $K \equiv 0$ on S .

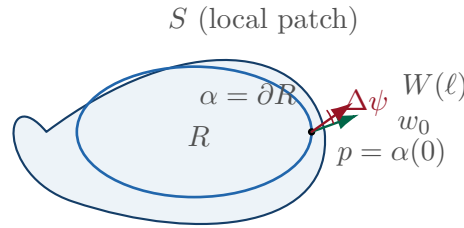


Figure 3: Loop $\alpha = \partial R$ on a surface patch: parallel transport around α returns a rotated vector; the net rotation is the holonomy angle $\Delta\psi$.

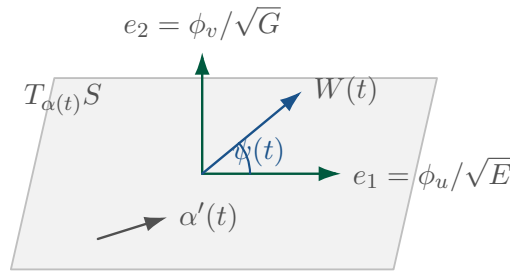


Figure 4: Local oriented orthonormal frame (e_1, e_2) and the angle coordinate $\psi(t)$ of the transported unit vector $W(t)$.

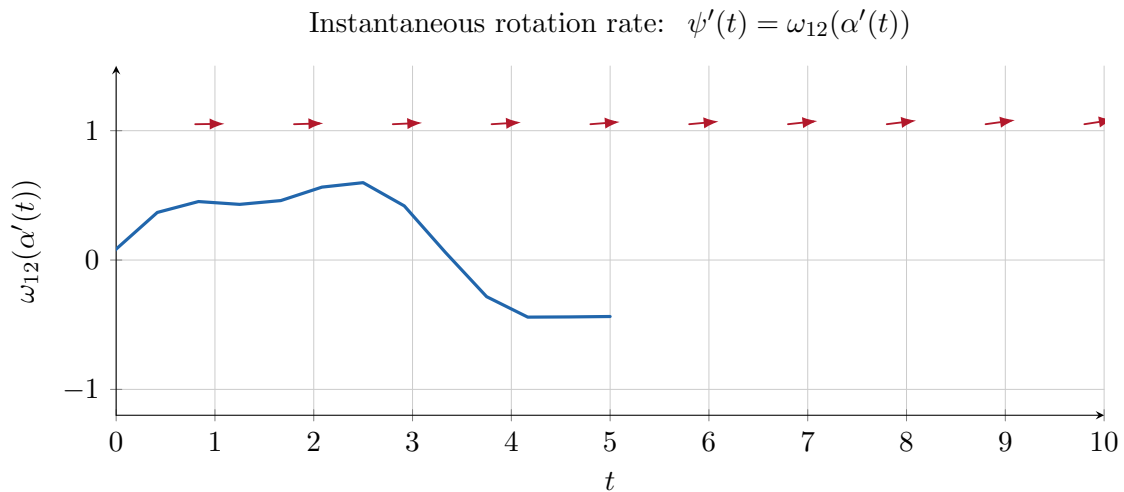


Figure 5: Along the curve, the connection form evaluated on $\alpha'(t)$ acts like an angular velocity. Integrating it yields the net holonomy angle.

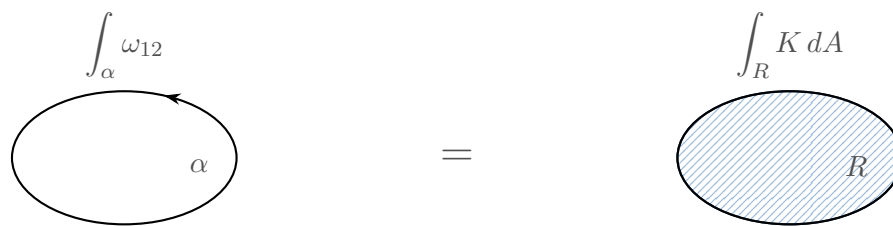


Figure 6: Stokes + structure equation: $\int_{\alpha} \omega_{12} = \int_R K dA$.

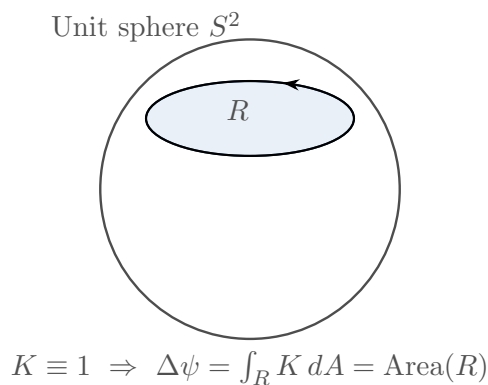


Figure 7: On the unit sphere, curvature is constant $K = 1$, so holonomy equals enclosed area (with consistent orientation conventions).



Figure 8: Both a plane and a round cylinder have $K = 0$. The cylinder is curved in $\mathbb{R}^3$ but intrinsically flat, so small-loop holonomy is 0.

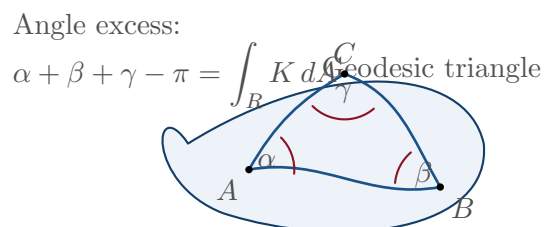


Figure 9: Geodesic triangle viewpoint: curvature produces angle excess; the defect is the integral of K over the enclosed region.

Problem 10 as definitions (parallel transport, holonomy, and curvature)

(1) Parallel transport (Levi–Civita). Let (S, g) be a Riemannian surface with Levi–Civita connection ∇ . For a piecewise smooth curve $\alpha : [0, \ell] \rightarrow S$, a vector field $W(t) \in T_{\alpha(t)}S$ is called *parallel* (along α) if

$$\nabla_{\alpha'(t)}W(t) = 0 \quad \text{for all } t.$$

Given $w_0 \in T_{\alpha(0)}S$, there exists a unique parallel field W with $W(0) = w_0$. The endpoint value defines the *parallel transport map*

$$P_\alpha : T_{\alpha(0)}S \rightarrow T_{\alpha(\ell)}S, \quad P_\alpha(w_0) = W(\ell).$$

It is linear and preserves the metric: $P_\alpha \in O(T_{\alpha(0)}S, T_{\alpha(\ell)}S)$, and if S is oriented, then for curves in an oriented chart $P_\alpha \in SO(2)$ (orientation preserving).

(2) Holonomy map and holonomy group. If $\gamma : [0, 1] \rightarrow S$ is a loop based at p (i.e. $\gamma(0) = \gamma(1) = p$), then parallel transport gives a linear isometry

$$P_\gamma : T_pS \rightarrow T_pS.$$

This is the *holonomy transformation* associated to γ . The *holonomy group* at p is

$$\text{Hol}_p(S, \nabla) = \{ P_\gamma \in GL(T_pS) : \gamma \text{ is a loop based at } p \}.$$

The *restricted holonomy group* is

$$\text{Hol}_p^0(S, \nabla) = \{ P_\gamma : \gamma \text{ is contractible (null-homotopic) in } S \}.$$

On an oriented surface, $\text{Hol}_p^0(S, \nabla) \subset SO(2)$, hence every P_γ is a rotation by some angle.

(3) Holonomy angle. Fix an oriented orthonormal basis (e_1, e_2) of T_pS . For any contractible loop γ based at p , the map $P_\gamma \in SO(2)$ is rotation by a unique angle $\Delta\psi(\gamma) \in \mathbb{R}$:

$$P_\gamma = \text{Rot}_{\Delta\psi(\gamma)}.$$

Equivalently, if $w_0 \in T_pS$ is a unit vector and W is its parallel transport along γ , then $\Delta\psi(\gamma)$ is the net change of the angle from e_1 to W after returning to p . (Choosing a different basis changes $\Delta\psi$ by multiples of 2π ; for small contractible loops one takes a continuous branch.)

(4) Connection 1-form (local definition). Let (e_1, e_2) be a local oriented orthonormal tangent frame on an open set $U \subset S$. The Levi–Civita connection is encoded by a 1-form ω_{12} on U defined by

$$\nabla_X e_1 = \omega_{12}(X) e_2, \quad \nabla_X e_2 = -\omega_{12}(X) e_1,$$

equivalently $\omega_{12}(X) = \langle \nabla_X e_1, e_2 \rangle$.

(5) Holonomy angle as an integral of the connection. If $\alpha : [0, \ell] \rightarrow U$ is a curve and W is a unit parallel field along α , writing

$$W(t) = \cos \psi(t) e_1(\alpha(t)) + \sin \psi(t) e_2(\alpha(t))$$

gives

$$\psi'(t) = \omega_{12}(\alpha'(t)), \quad \text{hence} \quad \Delta\psi(\alpha) = \psi(\ell) - \psi(0) = \int_{\alpha} \omega_{12}.$$

In particular, for a loop γ based at p contained in U ,

$$\Delta\psi(\gamma) = \int_{\gamma} \omega_{12}.$$

(6) Curvature 2-form and the structure equation. For an oriented orthonormal frame on a surface,

$$d\omega_{12} = K dA,$$

where K is the Gaussian curvature and dA is the Riemannian area form.

(7) Holonomy–curvature formula (Problem 10). Let $\alpha = \partial R$ be a positively oriented simple closed curve bounding a region $R \subset U$. Then by Stokes' theorem,

$$\Delta\psi(\alpha) = \int_{\alpha} \omega_{12} = \int_R d\omega_{12} = \int_R K dA.$$

This is exactly the statement of Problem 10:

$$\psi(\ell) - \psi(0) = \int_R K dA.$$

(8) Flatness criterion (path-independence). If parallel transport between any two points of a connected surface does not depend on the curve joining them, then every contractible loop has trivial holonomy: $\Delta\psi(\gamma) = 0$. Hence $\int_R K dA = 0$ for every sufficiently small region R , which implies $K \equiv 0$ on S .

Problem 11 (Curvature and torsion of a helix; consequences for closed curves)

Statement. If $a > 0$, calculate the curvature and torsion of the smooth curve

$$\alpha(s) = (a \cos(s/c), a \sin(s/c), bs/c), \quad c = \sqrt{a^2 + b^2}.$$

Suppose now that $\alpha : [0, 2\pi] \rightarrow \mathbb{R}^3$ is a smooth simple closed curve parametrized by arc-length with curvature everywhere positive.

- If both κ and τ are constant, show that $\kappa = 1$ and $\tau = 0$.
- If κ is constant and τ is not identically zero, show that $\kappa > 1$.
- If α is knotted and τ is constant, show that $\kappa(s) > 2$ for some $s \in [0, 2\pi]$.

Solution.

(A) The helix $\alpha(s) = (a \cos(s/c), a \sin(s/c), bs/c)$. Let $t = s/c$. Then $\alpha(s) = (a \cos t, a \sin t, bt)$ and $dt/ds = 1/c$. Compute

$$\alpha'(s) = \frac{1}{c} (-a \sin t, a \cos t, b), \quad \|\alpha'(s)\|^2 = \frac{1}{c^2}(a^2 + b^2) = 1,$$

so s is arc-length. Next,

$$\alpha''(s) = \frac{1}{c^2} (-a \cos t, -a \sin t, 0), \quad \|\alpha''(s)\| = \frac{a}{c^2}.$$

For a unit-speed curve, $\kappa = \|\alpha''\|$, hence

$$\boxed{\kappa = \frac{a}{c^2} = \frac{a}{a^2 + b^2}}.$$

To find τ , use the standard unit-speed formula

$$\tau = \frac{\det(\alpha', \alpha'', \alpha''')}{\|\alpha' \times \alpha''\|^2}.$$

A direct computation gives $\|\alpha' \times \alpha''\|^2 = \kappa^2$ and $\det(\alpha', \alpha'', \alpha''') = \frac{ab}{c^6}$, hence

$$\boxed{\tau = \frac{b}{c^2} = \frac{b}{a^2 + b^2}}.$$

(Equivalently: a circular helix has constant curvature $a/(a^2 + b^2)$ and constant torsion $b/(a^2 + b^2)$.)

(B) Closed unit-speed curves of length 2π with positive curvature.

(B1) If κ and τ are constant, then $\kappa = 1$ and $\tau = 0$. A unit-speed curve with constant curvature $\kappa > 0$ and constant torsion τ is, up to rigid motion, either a circle (if $\tau = 0$) or a circular helix (if $\tau \neq 0$). But a circular helix is not closed unless $\tau = 0$ (it has nonzero drift along its axis), so closedness forces $\tau = 0$.

Thus α is a circle of radius $1/\kappa$, hence its length is

$$L = \frac{2\pi}{\kappa}.$$

Since α is parametrized on $[0, 2\pi]$ by arc-length, $L = 2\pi$, so $\kappa = 1$. Therefore $\tau = 0$ and $\kappa = 1$.

(B2) If κ is constant and $\tau \neq 0$, then $\kappa > 1$. By Fenchel's theorem (total curvature),

$$\int_0^{2\pi} \kappa(s) ds \geq 2\pi,$$

with equality if and only if the curve is a convex planar curve. Here κ is constant and $L = 2\pi$, so the left-hand side is

$$\int_0^{2\pi} \kappa ds = \kappa \cdot 2\pi.$$

Hence $\kappa \geq 1$. If $\tau \neq 0$, then the curve is not planar, so equality cannot hold; therefore

$$\kappa \cdot 2\pi > 2\pi \quad \Rightarrow \quad \boxed{\kappa > 1}.$$

(B3) If α is knotted and τ is constant, then $\kappa(s) > 2$ for some s . By the Fáry–Milnor theorem, a knotted smooth closed curve satisfies

$$\int_0^{2\pi} \kappa(s) ds > 4\pi.$$

If, on the contrary, $\kappa(s) \leq 2$ for all $s \in [0, 2\pi]$, then

$$\int_0^{2\pi} \kappa(s) ds \leq \int_0^{2\pi} 2 ds = 4\pi,$$

contradicting the strict inequality above. Hence there exists s with

$$\boxed{\kappa(s) > 2}.$$

Examples.

- The curve in (A) is a circular helix; when $b = 0$ it reduces to a circle with $\kappa = 1/a$ and $\tau = 0$.
- A planar unit circle has $\kappa \equiv 1$, $\tau \equiv 0$, length 2π , matching (B1).
- A standard helix with $b \neq 0$ has $\tau \neq 0$ and is not closed, illustrating why (B1) forces $\tau = 0$.
- Any knotted curve (e.g. a trefoil) must have total curvature $> 4\pi$, forcing $\max \kappa > 2$ when $L = 2\pi$.

Theory for Problem 11: curvature, torsion, and total curvature of closed curves

1) Unit-speed curves and the Frenet frame. Let $\alpha : I \rightarrow \mathbb{R}^3$ be a C^3 curve parametrized by arc-length s (so $\|\alpha'(s)\| = 1$). Define the unit tangent

$$T(s) = \alpha'(s).$$

Assume the curvature $\kappa(s) = \|T'(s)\|$ is positive. Then the *principal normal* and *binormal* are

$$N(s) = \frac{T'(s)}{\|T'(s)\|} = \frac{T'(s)}{\kappa(s)}, \quad B(s) = T(s) \times N(s),$$

and (T, N, B) is the *Frenet frame*.

2) Frenet–Serret equations. For a unit-speed curve with $\kappa > 0$ one has

$$T' = \kappa N, \quad N' = -\kappa T + \tau B, \quad B' = -\tau N,$$

where $\tau(s)$ is the *torsion*. One convenient formula is

$$\tau(s) = -\langle B'(s), N(s) \rangle.$$

3) Curvature and torsion formulas in coordinates. For a general parameter t (not necessarily arc-length),

$$\kappa(t) = \frac{\|\alpha'(t) \times \alpha''(t)\|}{\|\alpha'(t)\|^3}, \quad \tau(t) = \frac{\det(\alpha'(t), \alpha''(t), \alpha'''(t))}{\|\alpha'(t) \times \alpha''(t)\|^2}.$$

If $\|\alpha'(s)\| = 1$, these simplify to

$$\kappa(s) = \|\alpha''(s)\|, \quad \tau(s) = \frac{\det(\alpha'(s), \alpha''(s), \alpha'''(s))}{\|\alpha'(s) \times \alpha''(s)\|^2}.$$

4) Classification: constant curvature and torsion. A C^3 unit-speed curve in $\mathbb{R}^3$ with constant $\kappa > 0$ and constant τ is (up to a rigid motion) either:

- a circle (planar case $\tau = 0$), or
- a circular helix (nonplanar case $\tau \neq 0$).

In particular, a circular helix has nonzero vertical drift and is not a closed curve.

5) Total curvature. For a C^2 curve, the *total curvature* is

$$\text{TC}(\alpha) = \int \kappa(s) ds$$

(over the whole curve; for a closed curve of length L , the integral is $\int_0^L \kappa(s) ds$).

6) Fenchel's theorem (closed curves). If α is a closed C^2 space curve, then

$$\int_0^L \kappa(s) ds \geq 2\pi,$$

with equality if and only if α is a convex planar curve (in particular, torsion $\tau \equiv 0$).

7) Fáry–Milnor theorem (knotted curves). If α is a knotted C^2 closed curve, then

$$\int_0^L \kappa(s) ds > 4\pi.$$

8) Easy consequences used in Problem 11. If α is closed, parametrized by arc-length, and has length $L = 2\pi$, then:

- If κ is constant, Fenchel gives $2\pi\kappa = \int_0^{2\pi} \kappa ds \geq 2\pi$, hence $\kappa \geq 1$, and if $\tau \neq 0$ (nonplanar), then $\kappa > 1$.
- If α is knotted, then $\int_0^{2\pi} \kappa ds > 4\pi$, so $\kappa(s) > 2$ for some s (otherwise $\kappa \leq 2$ everywhere would imply $\int_0^{2\pi} \kappa ds \leq 4\pi$).

Problem 12: (The Poincaré-Hopf theorem.)

Exact statement of the problem.

Let S be an oriented surface and $V : S \rightarrow \mathbb{R}^3$ a smooth vector field, that is, $V(p) \in T_p S$ for all $p \in S$. We say that p is *singular* if $V(p) = 0$. A singular point p is *isolated* if there exists a neighbourhood of p in which V has no other zeros. The singular point p is *non-degenerate* if

$$DV_p : T_p S \rightarrow T_p S$$

is a linear isomorphism (can you see why DV_p takes values in $T_p S$?). Show that if a singular point is non-degenerate, then it is isolated.

To each isolated singular point p we associate an integer called the *index* of the vector field at p as follows. Let $\phi : U \rightarrow S$ be an orthogonal parametrization around p compatible with the orientation. Let $\alpha : [0, l] \rightarrow \phi(U)$ be a regular piecewise smooth simple closed curve so that p is the only zero of V in the domain enclosed by α . Let $\varphi(t)$ be some differentiable determination of the angle from ϕ_u to $V(t) := V \circ \alpha(t)$. Since α is closed, there is an integer I (the index) defined by

$$2\pi I := \varphi(l) - \varphi(0).$$

1. Show that I is independent of the choice of parametrization (Hint: use an earlier problem). One can also show that I is independent of the choice of curve α , but this is a little harder. Also one can prove that if p is non-degenerate, then $I = 1$ if DV_p preserves orientation and $I = -1$ if DV_p reverses orientation.
2. Draw some pictures of vector fields in $\mathbb{R}^2$ with an isolated singularity at the origin. Compute their indices.
3. Suppose now that S is compact and that V is a smooth vector field with isolated singularities. Consider a triangulation of S such that
 - every triangle is contained in the image of some orthogonal parametrization;
 - every triangle contains at most one singular point;
 - the boundaries of the triangles contain no singular points and are positively oriented.

Show that

$$\sum_i I_i = \frac{1}{2\pi} \int_S K \, dA = \chi(S).$$

Thus, you have proved that the sum of the indices of a smooth vector field with isolated singularities on a compact surface is equal to the Euler characteristic (Poincaré-Hopf theorem). Conclude that a surface homeomorphic to S^2 cannot be combed.

Finally, suppose $f : S \rightarrow \mathbb{R}$ is a Morse function and consider the vector field given by the gradient of f , i.e., $\nabla f(p)$ is uniquely determined by $\langle \nabla f(p), v \rangle = Df_p(v)$ for all $v \in T_p S$. Use the Poincaré-Hopf theorem to show that $\chi(S)$ is the number of local maxima and minima minus the number of saddle points. Use this to find the Euler characteristic of a surface of genus two.

You can read more about this in Chap. VI of Milnor's 'Topology from the differential viewpoint'.

Problem 12: (The Poincaré–Hopf theorem.) — Solution

(A) A non-degenerate singular point is isolated. Let $\phi : U \subset \mathbb{R}^2 \rightarrow S$ be an orthogonal parametrization with $\phi(0) = p$. Write the tangent vector field in the local basis $\{\phi_u, \phi_v\}$:

$$V(\phi(u, v)) = a(u, v) \phi_u(u, v) + b(u, v) \phi_v(u, v).$$

Then $V(\phi(u, v)) = 0$ if and only if $F(u, v) := (a(u, v), b(u, v)) = (0, 0)$. Non-degeneracy of p means that

$$DV_p : T_p S \rightarrow T_p S$$

is an isomorphism; in these coordinates this is exactly the invertibility of $DF_{(0,0)}$ (the Jacobian matrix of (a, b) at $(0, 0)$). By the inverse function theorem, F is a local diffeomorphism near $(0, 0)$, so $F(u, v) = 0$ has the unique solution $(u, v) = (0, 0)$ in a neighbourhood. Hence p is an isolated singularity.

(i) Independence of the parametrization. Let $\alpha : [0, l] \rightarrow S$ be a regular piecewise smooth simple closed curve enclosing exactly one zero p of V , and assume $V(\alpha(t)) \neq 0$ for all t . Choose a continuous determination $\varphi(t)$ of the angle from ϕ_u to $V(\alpha(t))$, and define

$$I = \frac{\varphi(l) - \varphi(0)}{2\pi} \in \mathbb{Z}.$$

If $\tilde{\alpha} : [0, \tilde{l}] \rightarrow S$ is another parametrization of the same oriented curve, then $\tilde{\alpha} = \alpha \circ h$ for an increasing C^1 diffeomorphism $h : [0, \tilde{l}] \rightarrow [0, l]$ with $h(0) = 0$, $h(\tilde{l}) = l$. Setting $\tilde{\varphi}(t) := \varphi(h(t))$ gives a continuous angle determination for $V(\tilde{\alpha}(t))$, and

$$\tilde{\varphi}(\tilde{l}) - \tilde{\varphi}(0) = \varphi(l) - \varphi(0),$$

hence $\tilde{I} = I$. Therefore I does not depend on the parametrization.

Index of a non-degenerate zero. In local coordinates, $F(u) = DF_0 u + o(\|u\|)$. On a sufficiently small circle $\|u\| = \varepsilon$, the map $u \mapsto F(u)/\|F(u)\|$ has the same winding number (degree) as $u \mapsto DF_0 u / \|DF_0 u\|$. For an invertible linear map $A \in GL(2)$ this degree equals $\text{sign}(\det A)$. Thus, for a non-degenerate zero,

$$I(p) = \begin{cases} +1, & \det DF_0 > 0 \quad (\text{orientation preserving}), \\ -1, & \det DF_0 < 0 \quad (\text{orientation reversing}). \end{cases}$$

(ii) Examples in $\mathbb{R}^2$.

- Radial source/sink: $V(x, y) = (x, y)$ or $V(x, y) = -(x, y)$ has index $I = +1$.
- Saddle: $V(x, y) = (x, -y)$ has index $I = -1$.
- Higher index: identifying $\mathbb{R}^2 \cong \mathbb{C}$, $V(z) = z^k$ has index $I = k$.

(ii) Examples in $\mathbb{R}^2$ with an isolated singularity at the origin and their indices.

Throughout, the index at 0 is the winding number (degree) of the map

$$S^1 \ni (\cos t, \sin t) \mapsto \frac{V(\cos t, \sin t)}{\|V(\cos t, \sin t)\|} \in S^1,$$

equivalently

$$I = \frac{1}{2\pi} (\varphi(2\pi) - \varphi(0)),$$

where $\varphi(t)$ is a continuous determination of the angle of $V(\cos t, \sin t)$.

1. Source / sink (radial).

Consider

$$V(x, y) = (x, y) \quad \text{or} \quad V(x, y) = -(x, y).$$

On the unit circle $(x, y) = (\cos t, \sin t)$ we obtain

$$V(\cos t, \sin t) = (\cos t, \sin t) \quad \text{or} \quad V(\cos t, \sin t) = -(\cos t, \sin t) = (\cos(t+\pi), \sin(t+\pi)).$$

Thus the direction of V makes exactly one full turn as t goes from 0 to 2π :

$$\varphi(t) = t \quad \Rightarrow \quad \varphi(2\pi) - \varphi(0) = 2\pi,$$

and for the negative field

$$\varphi(t) = t + \pi \quad \Rightarrow \quad \varphi(2\pi) - \varphi(0) = 2\pi$$

as well. Hence in both cases

$$\boxed{I = +1}.$$

2. Saddle.

Consider

$$V(x, y) = (x, -y).$$

On the unit circle $(x, y) = (\cos t, \sin t)$:

$$V(\cos t, \sin t) = (\cos t, -\sin t) = (\cos(-t), \sin(-t)).$$

So a continuous angle choice is $\varphi(t) = -t$, and therefore

$$\varphi(2\pi) - \varphi(0) = -2\pi \quad \Rightarrow \quad \boxed{I = -1}.$$

3. Higher index (winding number k).

Identify $\mathbb{R}^2 \simeq \mathbb{C}$ via $z = x + iy$ and define

$$V(z) = z^k, \quad k \in \mathbb{Z}_{\geq 1}.$$

On the unit circle $z = e^{it}$ we get

$$V(e^{it}) = (e^{it})^k = e^{ikt}.$$

Thus $\varphi(t) = kt$ is a continuous determination of the argument, giving

$$\varphi(2\pi) - \varphi(0) = 2\pi k \quad \Rightarrow \quad \boxed{I = k}.$$

4. About index 0 (and a cautionary remark).

The following example is sometimes mistakenly cited as “index 0”, but it actually has index 2:

$$V(x, y) = (x^2 - y^2, 2xy).$$

Indeed, writing $z = x + iy$, we have

$$V(x, y) = (\Re(z^2), \Im(z^2)) \iff V(z) = z^2.$$

Hence by the previous item (with $k = 2$),

$$\boxed{I = 2}.$$

Remark. Index 0 *can* occur for an isolated singularity, but such zeros must be *degenerate* (in particular, they cannot be non-degenerate). Constructing an explicit smooth planar field with a *single* isolated zero of index 0 is possible but requires a more tailored construction; the key point is that the normalized direction map $S^1 \rightarrow S^1$ must have degree 0 (no net winding).

(iii) Poincaré–Hopf via triangulation. Let S be compact and V have isolated singularities p_i with indices I_i . Consider a triangulation satisfying the hypotheses of the problem. For each triangle T , define $I(T)$ as the winding number of the direction of V along ∂T (equivalently, the integer $\frac{1}{2\pi}(\Delta\varphi)_{\partial T}$). If T contains no singularity, then $I(T) = 0$; if T contains exactly one singularity p_i , then $I(T) = I_i$. Hence

$$\sum_i I_i = \sum_T I(T).$$

A Gauss–Bonnet type bookkeeping on each triangle gives

$$2\pi I(T) = \int_T K dA + (\text{exterior angle contributions}),$$

and when summed over all triangles the edge contributions cancel in pairs. The remaining combinatorics yields $2\pi\chi(S)$, so

$$\sum_i I_i = \frac{1}{2\pi} \int_S K dA = \chi(S).$$

In particular, on S^2 one has $\chi(S^2) = 2$, so a nowhere-zero tangent vector field cannot exist (“the sphere cannot be combed”).

Gradient of a Morse function. Let $f : S \rightarrow \mathbb{R}$ be Morse and take $V = \nabla f$. Then zeros of V are exactly critical points of f , and they are non-degenerate. At a critical point the index equals $\text{sign}(\det(\text{Hess } f))$: minima and maxima have index $+1$, saddles have index -1 . Therefore

$$\chi(S) = \sum_i I_i = (\#\text{min} + \#\text{max}) - (\#\text{saddles}).$$

For a closed orientable surface of genus 2,

$$\chi = 2 - 2g = 2 - 4 = -2.$$

Poincaré–Hopf / index explanation for the picture (periodic orbit obstruction)

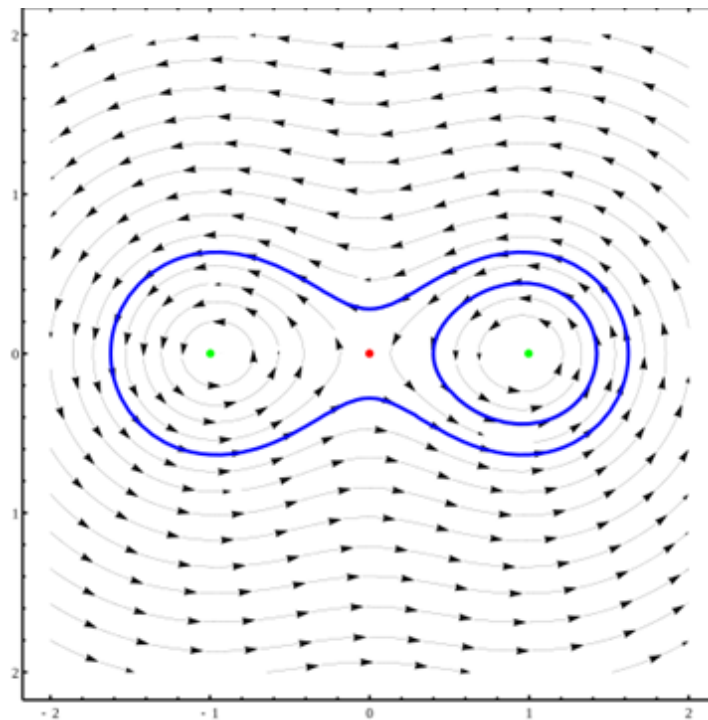


Figure 10: A planar vector field with closed trajectories (blue) surrounding equilibria: two centres (green) and one saddle (red). The index-counting rule implies a closed trajectory cannot enclose only a saddle.

What the picture shows. The black arrows represent a smooth planar vector field $V(x, y)$. The blue curves are *closed trajectories* (periodic orbits) of the flow. Inside the closed curves we see equilibrium points (zeros of V): two *centres* (green) and one *saddle* (red).

Index along a closed curve. Let $\gamma : S^1 \rightarrow \mathbb{R}^2$ be a positively oriented simple closed curve such that $V(\gamma(t)) \neq 0$ for all t . The *index of V along γ* is the winding number of the direction of the vector field along γ :

$$\text{Ind}_\gamma(V) = \frac{1}{2\pi} \Delta \arg(V(\gamma(t))) \in \mathbb{Z}.$$

A local form of the Poincaré–Hopf principle (or the degree theorem in the plane) states:

$$\text{Ind}_\gamma(V) = \sum_{p_i \text{ inside } \gamma} \text{Ind}_{p_i}(V),$$

where the sum runs over the isolated singularities (zeros) p_i of V in the region bounded by γ .

Indices of typical equilibria in $\mathbb{R}^2$. For an isolated *non-degenerate* equilibrium:

$$\text{Ind}_p(V) = \begin{cases} +1, & \text{centre / node / focus (orientation-preserving linearization),} \\ -1, & \text{saddle (orientation-reversing linearization).} \end{cases}$$

In particular, a centre has index $+1$ and a saddle has index -1 .

Why a periodic orbit forces boundary index $+1$. If γ is itself a *closed trajectory* of the flow, then V is tangent to γ everywhere. As one goes once around a positively oriented simple closed curve, the tangent direction rotates exactly once by 2π . Since V is everywhere tangent and never zero on γ , the direction of $V(\gamma(t))$ also makes exactly one full turn. Hence

$$\text{Ind}_\gamma(V) = +1.$$

The obstruction in the caption. Combining the two facts,

$$+1 = \text{Ind}_\gamma(V) = \sum_{p_i \text{ inside } \gamma} \text{Ind}_{p_i}(V).$$

Therefore the indices of the equilibria inside a periodic orbit must sum to $+1$.

- If γ encloses *one centre*, then the sum is $+1$, so this is possible.
- If γ encloses *two centres and one saddle*, the sum is

$$(+1) + (+1) + (-1) = +1,$$

so this is also possible (exactly as in the picture).

- If γ encloses *only one saddle*, the sum would be -1 , which cannot equal $+1$, so this configuration is impossible.

Why “Hamiltonian system” is mentioned. For planar Hamiltonian systems, trajectories lie on level sets of a Hamiltonian H , so closed trajectories naturally appear as closed level curves around centres, while saddles occur at “passes” connected by separatrices. The index-counting argument above explains why a closed level curve cannot surround just a saddle.

Can a curve enclose the saddle? Yes — but not a periodic orbit enclosing only a saddle.

It is important to distinguish:

1. **An arbitrary simple closed curve γ** in the plane (a loop you draw by hand).
2. **A closed trajectory (periodic orbit)** of the vector field V , i.e. a curve γ such that $\dot{\gamma}(t) = V(\gamma(t))$ (equivalently, V is everywhere tangent to γ and never vanishes on γ).

(A) Any closed curve can enclose a saddle. Geometrically/topologically, for any point p (including a saddle equilibrium) one can draw a small loop γ around p . There is no restriction here.

(B) A periodic orbit cannot enclose only a saddle. Assume γ is a positively oriented *periodic orbit* of V and that $V \neq 0$ on γ . Because V is tangent to γ , the direction of $V(\gamma(t))$ performs exactly one full turn as t runs once around the orbit. Hence the index of V along γ is

$$\text{Ind}_\gamma(V) = +1.$$

On the other hand, the planar index sum rule (a local Poincaré–Hopf/degree statement) gives

$$\text{Ind}_\gamma(V) = \sum_{p_i \text{ inside } \gamma} \text{Ind}_{p_i}(V).$$

A non-degenerate saddle has index $\text{Ind}_p(V) = -1$. Therefore, if γ enclosed *only* one saddle p , we would obtain

$$+1 = \text{Ind}_\gamma(V) = \sum_{p_i \text{ inside } \gamma} \text{Ind}_{p_i}(V) = \text{Ind}_p(V) = -1,$$

a contradiction. Thus:

A periodic orbit cannot surround exactly one saddle.

(C) A periodic orbit may enclose a saddle together with other equilibria. If additional singularities lie inside γ , the indices may sum to $+1$. For example:

$$(+1) + (+1) + (-1) = +1,$$

so a closed trajectory can enclose *two centres and one saddle* (as in the figure).

Hamiltonian remark. In planar Hamiltonian systems, trajectories lie on level sets $H = \text{const}$, so periodic orbits arise as regular closed level curves around centres. Saddles occur at critical points where separatrices appear. The index-counting argument above explains why a closed trajectory cannot enclose only a saddle.

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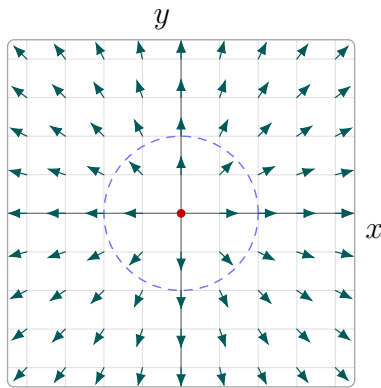
Problem 12(ii): Vector fields in $\mathbb{R}^2$ and their indices (gallery)

We draw several planar vector fields with an isolated singularity at the origin (marked in red). The dashed blue circle is a reference loop (e.g. the unit circle) used to visualize the winding of the vector direction. The index is the winding number of $V/\|V\|$ along a positively oriented simple closed curve around the singularity.

(1) **Radial source** $V(x, y) = (x, y)$

Index: $I=+1$

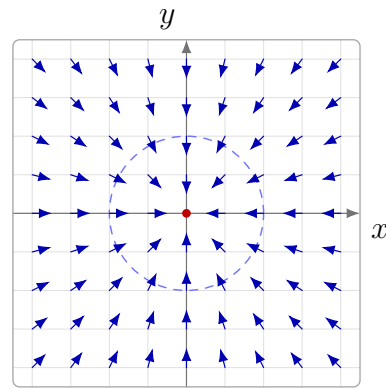
(one full positive turn)



(2) **Radial sink** $V(x, y) = (-x, -y)$

Index: $I=+1$

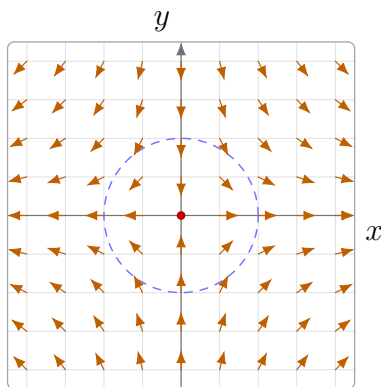
(direction still winds once)



(3) **Saddle** $V(x, y) = (x, -y)$

Index: $I=-1$

(one full negative turn)

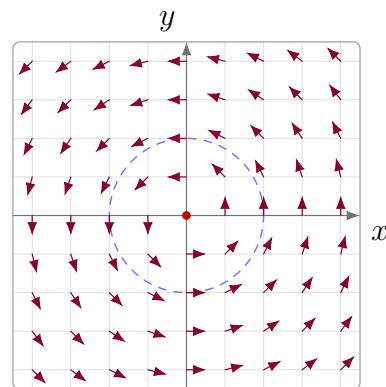


(4) **Center / rotation**

$V(x, y) = (-y, x)$

Index: $I=+1$

(closed orbits around the origin)

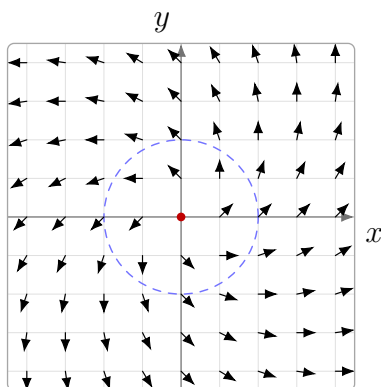


(5) **Spiral (focus)**

$V(x, y) = (x - y, x + y)$

Index: $I=+1$

(spiraling flow; same index as a node)



(6) **Double winding**

$V(x, y) = (x^2 - y^2, 2xy)$

Index: $I=2$

(equivalently $V(z) = z^2$ on $\mathbb{C}$)

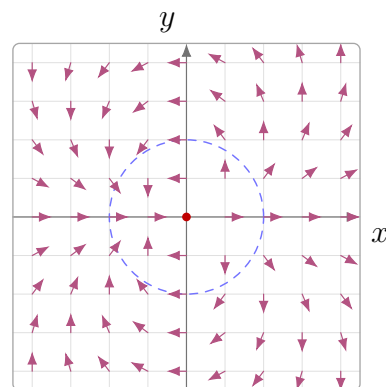


Figure 11: Six planar vector fields with an isolated singularity at the origin (red). The dashed circle is a reference loop used to read off the winding (index).

Problem 13: The degree of the Gauss map

Let S be a compact oriented surface and let $N : S \rightarrow S^2$ be the Gauss map. Consider $y \in S^2$ a regular value. Rather than counting their preimages modulo 2 as we did in the first lectures, we will count them with sign. Let $N^{-1}(y) = \{p_1, \dots, p_n\}$. Let $\varepsilon(p_i)$ be $+1$ if DN_{p_i} preserves orientation ($K(p_i) > 0$), and -1 if DN_{p_i} reverses orientation ($K(p_i) < 0$). Now let

$$\deg(N) := \sum_i \varepsilon(p_i).$$

As in the case of the degree mod 2, it can be shown that the sum on the right hand side is independent of the regular value and $\deg(N)$ turns out to be an invariant of the homotopy class of N .

Now, choose $y \in S^2$ such that y and $-y$ are regular values of N . Why can we do so? Let V be the vector field on S given by

$$V(p) := \langle y, N(p) \rangle N(p) - y.$$

1. Show that the index of V at a zero p_i is $+1$ if DN_{p_i} preserves orientation and -1 if DN_{p_i} reverses orientation.
2. Show that the sum of the indices of V equals twice the degree of N .
3. Show that $\deg(N) = \chi(S)/2$.

Problem 13: The degree of the Gauss map — Solution

Let S be a compact oriented surface and $N : S \rightarrow S^2$ its Gauss map. For a regular value $y \in S^2$, define

$$\deg(N) := \sum_{p \in N^{-1}(y)} \varepsilon(p), \quad \varepsilon(p) = \begin{cases} +1, & DN_p \text{ preserves orientation,} \\ -1, & DN_p \text{ reverses orientation.} \end{cases}$$

Choosing y so that y and $-y$ are regular. Let $\Sigma \subset S^2$ be the set of critical values of N . By Sard's theorem, Σ has measure 0 in S^2 . The antipodal map $A : S^2 \rightarrow S^2$, $A(y) = -y$, is a diffeomorphism, hence $A(\Sigma) = -\Sigma$ also has measure 0. Therefore $\Sigma \cup (-\Sigma)$ has measure 0, so we can choose

$$y \in S^2 \setminus (\Sigma \cup (-\Sigma)),$$

and then both y and $-y$ are regular values.

The vector field. Define

$$V(p) := \langle y, N(p) \rangle N(p) - y.$$

Decomposing y into normal and tangential parts at p ,

$$y = \langle y, N(p) \rangle N(p) + (y - \langle y, N(p) \rangle N(p)),$$

we obtain

$$V(p) = -(y - \langle y, N(p) \rangle N(p)),$$

so V is minus the tangential component of the constant vector y . Hence

$$V(p) = 0 \iff y \parallel N(p) \iff N(p) = \pm y,$$

so the zeros of V are exactly $N^{-1}(y) \cup N^{-1}(-y)$.

1. **Index at a zero.** Let p be a zero of V , so $N(p) = \pm y$. Differentiate $V = \langle y, N \rangle N - y$: for $w \in T_p S$,

$$DV_p(w) = \langle y, DN_p(w) \rangle N(p) + \langle y, N(p) \rangle DN_p(w).$$

Since $DN_p(w) \in T_{N(p)} S^2$ and $T_{N(p)} S^2 \perp N(p)$, we have $\langle N(p), DN_p(w) \rangle = 0$, and if $N(p) = \pm y$ then also $\langle y, DN_p(w) \rangle = 0$. Thus

$$\begin{cases} N(p) = y \Rightarrow \langle y, N(p) \rangle = 1 \Rightarrow DV_p = DN_p, \\ N(p) = -y \Rightarrow \langle y, N(p) \rangle = -1 \Rightarrow DV_p = -DN_p. \end{cases}$$

In dimension 2, $\det(-DN_p) = \det(DN_p)$, hence

$$\text{sign det}(DV_p) = \text{sign det}(DN_p).$$

For a nondegenerate zero, the local index satisfies $\text{ind}_p(V) = \text{sign det}(DV_p)$, so

$$\text{ind}_p(V) = +1 \iff DN_p \text{ preserves orientation,} \quad \text{ind}_p(V) = -1 \iff DN_p \text{ reverses orientation}$$

(Equivalently, $\text{sign det}(DN_p) = \text{sign } K(p)$.)

2. **Sum of indices equals $2 \deg(N)$.** Since the zeros are $N^{-1}(y) \cup N^{-1}(-y)$,

$$\sum_{V(p)=0} \text{ind}_p(V) = \sum_{p \in N^{-1}(y)} \varepsilon_y(p) + \sum_{p \in N^{-1}(-y)} \varepsilon_{-y}(p),$$

where $\varepsilon_{\pm y}(p)$ is the orientation sign of $DN_p: T_p S \rightarrow T_{\pm y} S^2$. Each sum is the signed degree of N computed at the regular value $\pm y$, hence

$$\sum_{p \in N^{-1}(y)} \varepsilon_y(p) = \deg(N) = \sum_{p \in N^{-1}(-y)} \varepsilon_{-y}(p),$$

so

$$\sum_{V(p)=0} \text{ind}_p(V) = 2 \deg(N).$$

3. **Conclusion** $\deg(N) = \chi(S)/2$. By the Poincaré–Hopf theorem,

$$\sum_{V(p)=0} \text{ind}_p(V) = \chi(S).$$

Combining with (ii) gives

$$\chi(S) = 2 \deg(N) \implies \deg(N) = \frac{\chi(S)}{2}.$$

Examples

- **Sphere S^2 :** $N = \text{id}$, so $\deg(N) = 1$ and $\chi(S^2) = 2$, hence $1 = \chi/2$.
- **Torus T^2 :** $\chi(T^2) = 0$, hence $\deg(N) = 0$.
- **Genus g surface:** $\chi = 2 - 2g$, hence $\deg(N) = \frac{2-2g}{2} = 1 - g$.

Degree mod 2 vs. signed degree (orientation count)

Let $f: M^n \rightarrow N^n$ be a smooth map between compact n -manifolds, and let $y \in N$ be a *regular value*. Then $f^{-1}(y)$ is a finite set.

Degree mod 2. The mod-2 degree keeps only the *parity* of the number of preimages:

$$\deg_2(f) := \#f^{-1}(y) \pmod{2} \in \{0, 1\}.$$

This definition does *not* use orientations. It forgets all sign information: any two points contribute $2 \equiv 0 \pmod{2}$, regardless of how Df behaves.

Signed degree (integer degree). Assume now that M and N are oriented. For each $p \in f^{-1}(y)$, the differential

$$Df_p: T_p M \rightarrow T_y N$$

is an isomorphism. Define the *local sign*

$$\varepsilon(p) := \begin{cases} +1, & \text{if } Df_p \text{ preserves orientation,} \\ -1, & \text{if } Df_p \text{ reverses orientation.} \end{cases}$$

Then the (signed) degree is

$$\deg(f) := \sum_{p \in f^{-1}(y)} \varepsilon(p) \in \mathbb{Z}.$$

Thus $\deg(f)$ counts *orientation-preserving hits minus orientation-reversing hits*.

Relation between the two. Reducing the signed degree modulo 2 recovers the mod-2 degree:

$$\deg(f) \equiv \#f^{-1}(y) \pmod{2},$$

since $-1 \equiv +1 \pmod{2}$. In other words, mod-2 degree is the signed degree with the signs forgotten.

Where the “two directions” appear for the Gauss map

For an oriented surface $S \subset \mathbb{R}^3$, the Gauss map is a continuous choice of unit normal

$$N: S \rightarrow S^2.$$

At each point $p \in S$ there are *two* unit normals, $N(p)$ and $-N(p)$, but the *orientation of S* selects one of them continuously; this is exactly what makes N well-defined.

In Problem 13 one chooses $y \in S^2$ so that both y and $-y$ are regular values. The sets

$$N^{-1}(y) \quad \text{and} \quad N^{-1}(-y)$$

are generally different: they correspond to points where the chosen normal points in the y -direction or in the opposite direction.

The vector field

$$V(p) := \langle y, N(p) \rangle N(p) - y$$

has zeros precisely at points where $N(p) = \pm y$; hence its index sum captures contributions from *both* directions on the sphere. This is the geometric reason a factor of 2 appears later (the total index sum becomes $2 \deg(N)$).

Which orientation is being checked?

Let $f: S \rightarrow S^2$ be a smooth map between oriented surfaces, and let $y \in S^2$ be a regular value. For each $p \in f^{-1}(y)$ the differential

$$Df_p: T_p S \longrightarrow T_y S^2$$

is a linear isomorphism.

Orientation preservation is a comparison between *both* sides. The sign $\varepsilon(p) \in \{+1, -1\}$ is defined by comparing the given orientation on the *domain tangent plane* $T_p S$ with the given (standard) orientation on the *target tangent plane* $T_y S^2$:

$$\varepsilon(p) = \begin{cases} +1, & \text{if } Df_p \text{ sends positively oriented bases of } T_p S \text{ to positively oriented bases of } T_y S^2, \\ -1, & \text{if } Df_p \text{ sends positively oriented bases of } T_p S \text{ to negatively oriented bases of } T_y S^2. \end{cases}$$

Equivalently, if A is the matrix of Df_p written in any positively oriented basis of $T_p S$ and any positively oriented basis of $T_y S^2$, then

$$\varepsilon(p) = \text{sign}(\det(A)) = \begin{cases} +1, & \det(A) > 0, \\ -1, & \det(A) < 0. \end{cases}$$

Thus one is not checking orientation on S alone or on S^2 alone; one is checking whether the *map between the two oriented tangent planes* preserves or reverses orientation.

Gauss map case. For the Gauss map $N: S \rightarrow S^2$, at a point p with $y = N(p)$ a regular value, DN_p is an isomorphism and one has (up to the conventional sign choice for the shape operator)

$$\det(DN_p) = K(p),$$

so $\varepsilon(p) = +1$ when $K(p) > 0$ and $\varepsilon(p) = -1$ when $K(p) < 0$.